

Understanding the metrics of Internet broadband access: How much is enough?

David D. Clark

Sara C. Wedeman

MIT

Version of 9/12/2022

1. Introduction

Typical offerings of wireline broadband access are tiered according to the maximum download and upload speeds the user will be able to achieve. Speed tiers have emerged as an approach to segment the market, but leave consumers with a puzzle: what speed tier will be sufficient for their needs? As well, there are other measures of broadband quality that are now getting increasing attention, including various measures of latency and packet loss rates.

The goal of this paper is to provide some practical insights about how to interpret these measures: how much speed is enough, how little latency is good enough, and what packet loss rates will impair performance?

These questions are gaining importance for at least three reasons.

First, a recent law in the U.S. requires the FCC to define a label, in a standard format, that ISPs must use to describe the pricing and performance metrics of their various offerings. The analogy is to the nutrition labels required on food products. We can assume that metrics such as speed, latency and loss rate will be required on that label. But how is the consumer to make sense of the numbers so reported?

Second, as public moneys are being used to subsidize the construction of broadband access networks in unserved areas, the government must determine what sorts of performance metrics should be required of the networks that are proposed. In this case it is not the consumer that must make sense of the numbers, but the agencies responsible for evaluating and authorizing different proposals. Should the same metrics be used in both cases?

Third, as sufficient access speeds become less of a limitation for many users, other metrics such as latency are becoming more important, and both consumers and regulators need to focus on these other metrics.

This paper is organized as follows. We begin with some general comments about the idea of a “broadband nutrition label” and then use this as an opening to explore the questions above: how much (or little) is enough. We first frame our answers through the lens of consumer understanding—how can we provide information in a form that will attract the attention of and be comprehensible to consumers at all levels of technology awareness. We note the circumstances where a regulator might benefit from (or require) a different form of answer than is given to the consumer

2. Constructing a “broadband nutrition label”

Since the concept of broadband nutrition labels is analogous to that of actual nutrition labels, which were instituted when the Nutrition Labeling and Education Act (NLEA)¹ was signed into law in November 1990, it seems worthwhile to consider whether and how such an approach could be effective at helping consumers feel confident in choosing between competing broadband access service levels and providers. We would argue that many consumers find technology intimidating, and that the level of

¹ Waxman, H. A. (1990, August 11). H.R.3562 - 101st Congress (1989-1990): Nutrition Labeling and Education Act of 1990. Retrieved from <https://www.congress.gov/bill/101st-congress/house-bill/3562>

knowledge required to determine whether a particular offering is a 'good deal' may require more technical knowledge than one needs to make an informed choice between two competing brands of peanut butter. This makes it particularly important to ensure that the labels a) attract their attention, and b) are comprehensible, and c) present meaningful, realistic choices.

There is a large body of research on the efficacy of actual nutrition labels in supporting fact-based decision-making, with results covering everything from color and font size to the placement of the label.^{2,3} From the fields of cognitive psychology and behavioral economics, we have learned - among other things - that in general, people have difficulty making decisions based on information that is complicated, where they lack the knowledge necessary to ascertain the meaning of the inputs, and based on non-substantive elements of presentation (e.g. numbering, placement on a list, familiarity).^{4,5} In "Psychology, Behavioral Economics, and Public Policy," the authors show how the complexity of choosing a 401K plan may cause people to simply avoid making a decision at all - thus losing out on a financial benefit that, over time, may bring them great value.⁶

These cautionary comments are intended to emphasize that in providing consumer-facing information, simplicity, meaningfulness, and ease of understanding are primary concerns. As well, to be credible, the information provided by the ISP must be independently verifiable. This is a key requirement; in that it offers an empirical basis for confidence in the accuracy of the label.

How would we expect a consumer to use a "broadband nutrition label"? The performance numbers in the broadband nutrition label can serve two purposes. One is simply to identify the service being described—"this is the 400 mb/s service", for example. The other purpose is to help consumers pick among the services to select the one that best matches their needs.

For the consumer to use these numbers to pick among services, there must be some context that makes the numbers meaningful. The original food nutrition label, as specified by the FDA and written into the law, includes an important piece of information, the Daily Value or DV (previously called the Minimum Daily Requirement). Each nutritional value is presented along with the percent of the DV the serving provides. Figure 1 shows a sample broadband label as proposed by the FCC in 2016. The section on performance lists four measures, but has no analog to the DV, and thus provides no guidance to consumers about the meaning of the reported values.

We argue that a broadband nutrition label must include an analog to the DV, which we call the Satisfactory Service Level or SSL. In the next sections of this paper, we provide background on how an SSL might be specified for each of the performance parameters in the proposed label, and we suggest a specific interpretation of the parameter (a definition/measurement method) which we believe strikes the best balance between the goals of simplicity, meaningfulness, and independent verifiability.

² Van den Wijngaart, Annoek W E M. (2002): "Nutrition labelling: purpose, scientific issues and challenges." Asia Pacific journal of Clinical Nutrition Vol. 11,2 S68-71. doi:10.1046/j.1440-6047.2002.00001.

³ Talati, Z., Pettigrew, S. et al. (2017) "The relative ability of different front-of-pack labels to assist consumers discriminate between healthy, moderately healthy, and unhealthy foods" *Food Quality and Preference*, Vol. 59 p 109-113, <https://doi.org/10.1016/j.foodqual.2017.02.010>.

⁴ Irrational Labs (2022) "16 critical cognitive biases (plus key academic research)" Retrieved from <https://irrationallabs.com/blog/cognitive-biases-and-academic-research>

⁵ Blakstad, Sofie, and Robert Allen. "Decision Architecture." *FinTech Revolution*. Palgrave Macmillan, Cham, 2018. 389-401.

⁶ Amir, O., Ariely, D., Cooke, A. et al. Psychology, Behavioral Economics, and Public Policy. *Market Lett* **16**, 443–454 (2005). <https://doi.org/10.1007/s11002-005-5904-2>

Many believe, and some live by, the notion that “you get what you measure.”⁷ While rarely reported in its entirety, the author of the phrase, V.F. Ridgeway, actually said:

*What gets measured gets managed - even when it's pointless to measure and manage it, and even if it harms the purpose of the organisation to do so.*⁸

We believe this insight offers essential guidance as the Commission develops its approach to launching any 'nutrition label'- type program in the telecommunications arena. As the FCC identifies values to put into a nutrition label, it should remember that the metrics it designates will be the metrics providers optimize. To truly benefit the consumer, characteristics measured should map in some understandable way to a good user experience.

To illustrate this point, imagine that cars were marketed based on their top speeds. Manufacturers would then strive to produce cars that could go faster than those made by competitors. “Truth in advertising” testing would follow, designed to determine which automobiles were the fastest. Setting aside the explosion in automobile accidents, many consumers would realize that comparing cars on this basis had nothing to do with the quality of their driving experience. Today the performance of cars is characterized using fuel economy, the time to accelerate from 0-60 mph, protection in crashes, and the like, and most drivers have at least some intuition about how to interpret these measures.

If one decides to measure top speed, the result is cars that go uselessly fast. If the FCC decides to include download speed in their nutrition label, they must accompany this with some interpretive information, both to help consumers pick the right value and to avoid a ‘more is better’ speed race and the potential for unjustified upselling that can lead to unnecessary expenditures for consumers—the exact opposite of the goal of the label.

Once we choose performance parameters to characterize broadband access (for example: speed), there will then be a focus on the “truth in advertising” question—are the ISPs providing what they promised? While a focus on truth in advertising is going to be a reality in the marketplace, it has nothing to do with ‘how much is enough,’ and can distract from that question, which is the question that should matter to the consumer when picking among services.

The numbers that are included in a nutrition label should not be the result of actual measurement—they should be target or nominal values the consumer can expect to achieve. A proposal to use actual performance testing as a basis for the values in a nutritional label will raise many issues—how is the testing standardized, how frequently would the testing (and the resulting values in the label) be revised, and how can a new service offering come to market before there is a deployed base to test? The use of a target value as the published number will be easiest to specify when there is a prior understanding of what that target value might mean, as with download and upload speeds. However, for metrics such as latency and packet loss rate, it will be necessary to define what a target value might mean (and how subsequent “truth in advertising” testing can be performed), as part of adding that measure to the nutrition label.

⁷ Buerkli, D. (2021). “What gets measured gets managed — It’s wrong and Drucker never said it.” Medium. Retrieved from <https://medium.com/centre-for-public-impact/what-gets-measured-gets-managed-its-wrong-and-drucker-never-said-it-fe95886d3df6>

⁸ Ridgeway, V. F. (1956). "Dysfunctional Consequences of Performance Measurements." *Administrative Science Quarterly*, 1(2), 240–247. <https://doi.org/10.2307/2390989>

Broadband Facts	
Fixed broadband consumer disclosure	
Choose Your Service Data Plan for 50Mbps Service Tier	
Monthly charge for month-to-month plan	\$60.00
Monthly charge for 2 year contract plan	\$55.00
Click here for other pricing options including promotions and options bundled with other services, like cable television and wireless services.	
Other Charges and Terms	
Data included with monthly charge	300GB
Charges for additional data usage – each additional 50GB	\$10.00
Optional modem or gateway lease – Customers may use their own modem or gateway; click here for our policy	\$10.00/month
Other monthly fees	Not Applicable
One-time fees	
Activation fee	\$50.00
Deposit	\$50.00
Installation fee	\$25.00
Early termination fee	\$240.00
Government Taxes and Other Government-Related Fees May Apply: Varies by location	
Other services on network	
Performance - Individual experience may vary	
Typical speed downstream	53 Mbps
Typical speed upstream	6 Mbps
Typical latency	35 milliseconds
Typical packet loss	0.08%
Network Management	
Application-specific network management practices?	Yes
Subscriber-triggered network management practices?	Yes
More details on network management	
Privacy	See our privacy policy
Complaints or Inquiries	To contact us: online /(123)456-7890; To submit complaints to the FCC: online /(888)225-5322
Learn more about the terms used on this form and other relevant information at the FCC's website.	

Figure 1: A sample broadband nutrition label, from the FCC in 2016. Note that the performance section does not include any analog to a Daily Value—any guidance as to how to interpret these numbers.⁹

⁹ <https://www.fcc.gov/broadbandlabels>

3. Understanding the relevance of a performance number.

What would it mean for a performance parameter to be meaningful - to be a viable analogue to the DV; an indicator of relevance and substance? As we begin to address this question, we introduce two important terms: QoS and QoE.

The network community distinguishes between measures of *Quality of Service* (QoS), which comprises a set of technical measures of network performance, and *Quality of Experience* (QoE), which focuses on user satisfaction with the experience of using various Internet applications. When looking at QoS parameters, the appropriate question is: what combination of characteristics, at what levels, create a satisfactory user experience. The answer to this question can only be derived from an understanding of how QoS parameters relate to QoE.

The question of how to operationalize the concept, "adequate broadband performance," has been posed frequently for most of the past twenty years. In 2002, the Computer Science and Telecommunications Board of the National Academies was asked to study and report on the challenges and opportunities of getting broadband Internet access to U.S. residential users.¹⁰ As part of this effort, the committee was asked to specify what levels of Internet access and performance would be sufficient for most. The committee refused to provide a numeric answer, because it understood that the answer would change over time. Instead, the group proposed two criteria for answering the question: 'how much is enough':

- *Broadband Definition 1. Local access link performance should not be the limiting factor in a user's capability for running today's applications.*
- *Broadband Definition 2. Broadband services should provide sufficient performance—and wide enough penetration of services reaching that performance level—to encourage the development of new applications.*

These definitions clarified that the basis for determining "how much is enough?" was the relationship between the needs of applications being used by residential customers (the resulting QoE) and the extant capacity of the Internet.

4. Download speeds.

To introduce the challenge of making measurements meaningful while keeping them simple, we discuss in relative detail the specification of download speed, which is one of the easiest to understand of the various parameters in the proposed label.

While some consumers may have a reasonable understanding of the meaning of download speeds - what they are and are not - we suspect that many users have little idea what speeds are required to support common applications with good QoE. At the same time, speed, at first blush, seems like a simple number to understand. We believe that many contemporary Internet marketing programs stress speed, in part, because it is superficially comprehensible. However, this emphasis advances the notion that 'more is better.' In turn, this fosters a dynamic in which ISPs can "upsell" a consumer to a higher speed tier that does not improve the customer's QoE and primarily benefits the provider's bottom line.

¹⁰ National Research Council. Broadband: Bringing Home the Bits. The National Academies Press, Washington, DC, 2002

Variation in consumer needs.

One of the key differences from the DV in nutrition and an SSL in broadband services is that the FDA, in pursuit of simplicity, has specified a single value for the DV, appropriate for the nutritional needs of a “typical adult,”¹¹ ignoring the real issue that different people may have different requirements. With respect to broadband, people from different walks of life have different broadband needs. Some users may have multiple high-resolution televisions or multiple computers in constant use; other users may focus on home-based work or schooling; others may be passionate players of online games, and so on. Given the likely variety among user types, we recommend that the guidelines include typical usage profiles (large family, small family, range of using habits, etc.) against which consumers can compare themselves to get some understanding of the most relevant SSL for them.¹²

SSLs for download speeds

Many sources give estimates of necessary bandwidth for various applications. For applications that involve streaming video, the actual data rates for various encoding are well-known. However, the data rates depend on the resolution (e.g., SD at 480p, or HD at 720p or 1080p), the frame rate (24 frames/second to 60 fps), the complexity of the scene, and depth of color encoding. Many sources give what they consider to be a typical answer without specifying all those parameters. As well, the capacity of the service from the ISP being used to stream video should be provisioned to allow for fluctuations in actual performance. A realistic approach would be to add a “headroom factor” of perhaps 33% to 50% when selecting a service. Some sites include a recommended headroom, others give the actual data rate.

Some common video encoding and the range of required speeds are as follows:

- SD (480p): .8-1.2 mb/s
- HD (720p): 1.2-1.9 mb/s
- HD (1080p, 60 fps): 1.9-5.9 mb/s
- 4k (60 fps): 18-25 mb/s.

Using these numbers, one can propose a somewhat simple SSL for download speed by noting that for each simultaneous HD stream, no more than 5.9 mb/s*1.5 headroom or about 9 mb/s is necessary. One could probably watch 3 HD streams at once over a 25 mb/s service.

Streaming video can be embedded in a number of applications, so these numbers might apply to watching traditional linear television, YouTube, videos embedded in Facebook, and so on. This leads to very different estimates of required speeds for those apps, depending on whether they assume the consumer is watching video.

Teleconferencing also transmits encoded video, but usually at lower resolutions, so the required bandwidths are lower. Zoom provides the following advice:¹³

For 1:1 video calling:

- For high-quality video: 600kbps (up/down)
- For 720p HD video: 1.2Mbps (up/down)
- For 1080p HD video: 3.8Mbps/3.0Mbps (up/down)

For group video calling:

- For high-quality video: 1.0 Mbps/600kbps (up/down)
- For 720p HD video: 2.6Mbps/1.8Mbps (up/down)

¹¹ Specifically, a human for whom the required daily caloric intake is 2,000 calories.

¹² If it has not been done already, the FCC should institute a formal program of consumer research, otherwise the segments it strives to serve will range from “someone and something” to “nobody and no known thing,” and back.

¹³ <https://support.zoom.us/hc/en-us/articles/201362023-System-Requirements-for-PC-Mac-and-Linux#bandwidth>

- For 1080p HD video: 3.8Mbps/3.0Mbps (up/down)
- For gallery view receiving: 2.0Mbps (25 views), 4.0Mbps (49 views)

The higher resolutions mentioned by zoom are used only if they are specifically enabled when the meeting is set up (for example to allow a high-quality recording of the meeting) so typical speeds for Zoom are under 2 mb/s.

The other application that has significant speed requirements is the web, and here the numbers are even more variable, because there is no hard threshold to carry a given encoding. The faster the download speed, the faster the web load time, up to a point. However, at some point, additional speed does not improve the QoE, because the limiting factor becomes the latency to the sources of the content. We used data from the FCC Measuring Broadband America (MBA) program to illustrate this point. The MBA program measures web loading time, for different advertised speed tiers. An analysis of the MBA web load time data is shown in Figures 2a and 2b.

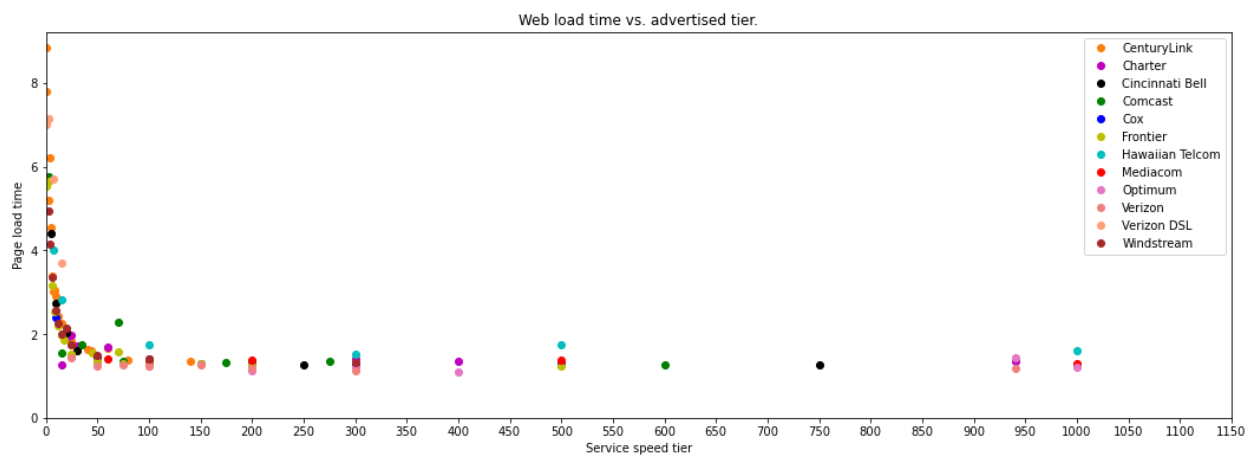


Figure 2a: Plot of web page load time, vs. speed tier for the test, as measured by the FCC Measuring Broadband America program in 2020.¹⁴ Points are median load time across all measurement units for each provider at each tier. Above a certain speed, more bandwidth does not reduce the load time. The minimum load time never drops below about a second. Above those tiers, the limit on the page load time is the latency of the transfers required to load the page.

¹⁴ Plot derived from validated data from 11th Fixed MBA Report, released December 31, 2021.
<https://www.fcc.gov/reports-research/reports/measuring-broadband-america/validated-data-measuring-fixed-broadband-0>

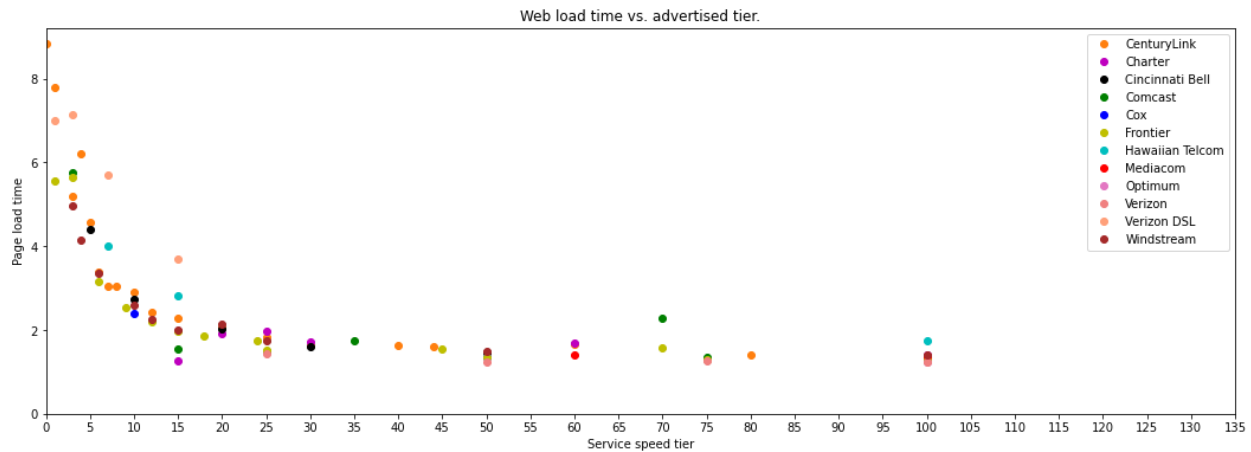


Figure 2b: To make it easier to see where the knee of the curve is in Figure 2a, this figure omits the data points above 100 mb/s. Visually, speed tiers above about 20 mb/s do not seem to offer an improvement, so an SSL of 15 mbps would be sufficient for typical web browsing.

Above about 20 mb/s, adding more speed does not improve the load time. The limit on the load time is the latency to the servers providing the elements of the web page.

It is interesting to see how this plot has changed over time. Figure 3 shows a similar analysis of MBA page load time, from 2013.

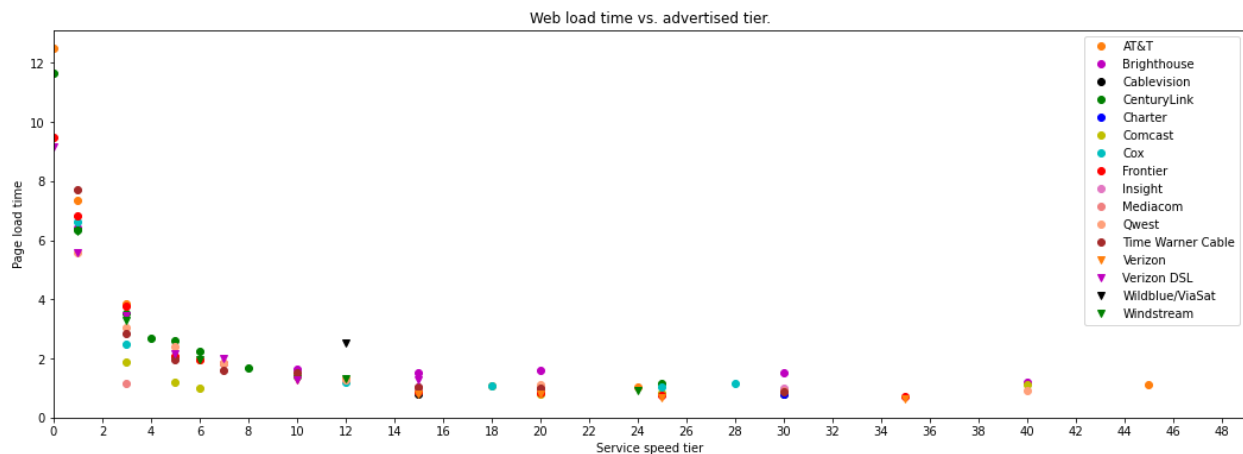


Figure 3. Plot of web loading time from MBA data, vs. advertised speed of the access technology in 2013. For all technologies, above about 10 or 12 mb/s, there is not a substantial reduction in page load time, so a simple SSL for web loading as of 2013 would be between 10 and 12 mb/s.

Figure 3 shows that in 2013, above 10 or 12 mb/s, more download speed would not have further reduced the total load time.

For a historical comparison, Figure 4 shows a similar plot, computed in 1999. At that time, most users had 56kb dialup, web pages were much simpler, and the key of the curve is close to 1 mbps.

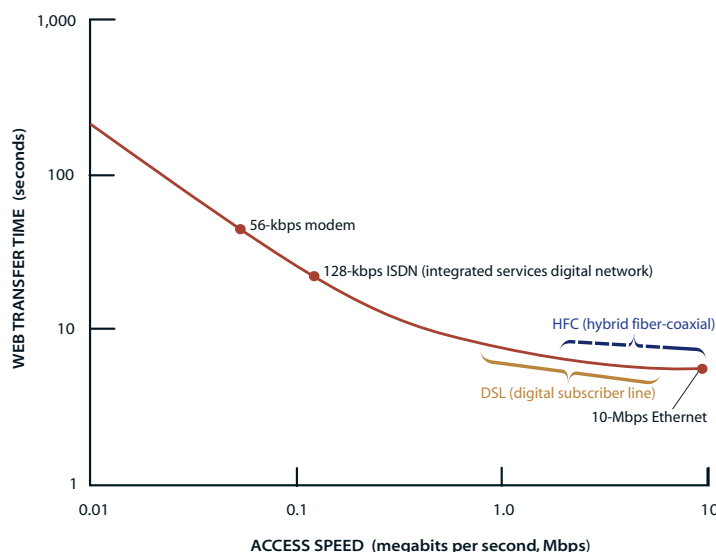


Figure 4: Page load time vs. access speed, as of 1999. Typical users had 56 kb modems, HFC (cable modem technology) had a maximum speed of around 10 mbps, but residential broadband was not commonplace. At the time, it seemed as if DSL might be the most common solution.¹⁵

Figure 4 makes the point that applications evolve in response to changes in the underlying technology. The report from the National Research Council (2002) we quoted above was written 3 years after this plot was computed, and at the time, access speed sufficient to meet the needs of then-current applications was perhaps 1 or 2 mbps.

Some examples showing how information about an SSL could be presented

Once we agree on values for an SLL, what is the best way to present these to a consumer? Various organization, including the FCC, have proposed usage profiles that illustrate what an SSL for download speed might look like. Here are four examples, presenting the information as a table, a graphic, a plot, and a narrative:

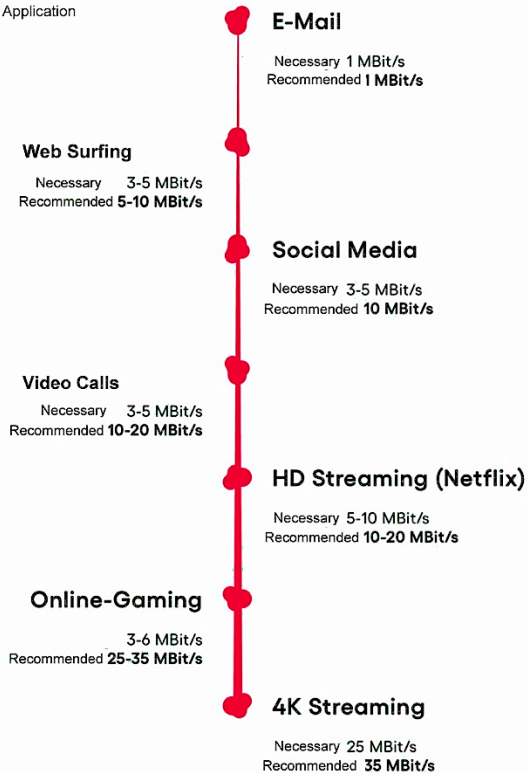
¹⁵ Plot prepared by Xiaowei Yang, then at MIT. Access speeds were controlled by a rate-limiting device in the path of a machine running the test client. Clark, David. "High-speed Data Races Home," Scientific American, October 1999, Vol 284, Num 4.

Activity	Minimum Download Speed (Mbps)
General Usage	
General Browsing and Email	1
Streaming Online Radio	Less than 0.5
VoIP Calls	Less than 0.5
Student	5 - 25
Telecommuting	5 - 25
File Downloading	10
Social Media	1
Watching Video	
Streaming Standard Definition Video	3 - 4
Streaming High Definition (HD) Video	5 - 8
Streaming Ultra HD 4K Video	25
Video Conferencing	
Standard Personal Video Call (e.g., Skype)	1
HD Personal Video Call (e.g., Skype)	1.5
HD Video Teleconferencing	6
Gaming	
Game Console Connecting to the Internet	3
Online Multiplayer	4

Example 1: FCC Broadband Speed Guide ¹⁶, showing SSL information as a table. This table does not discuss that there might be multiple users sharing an access link.

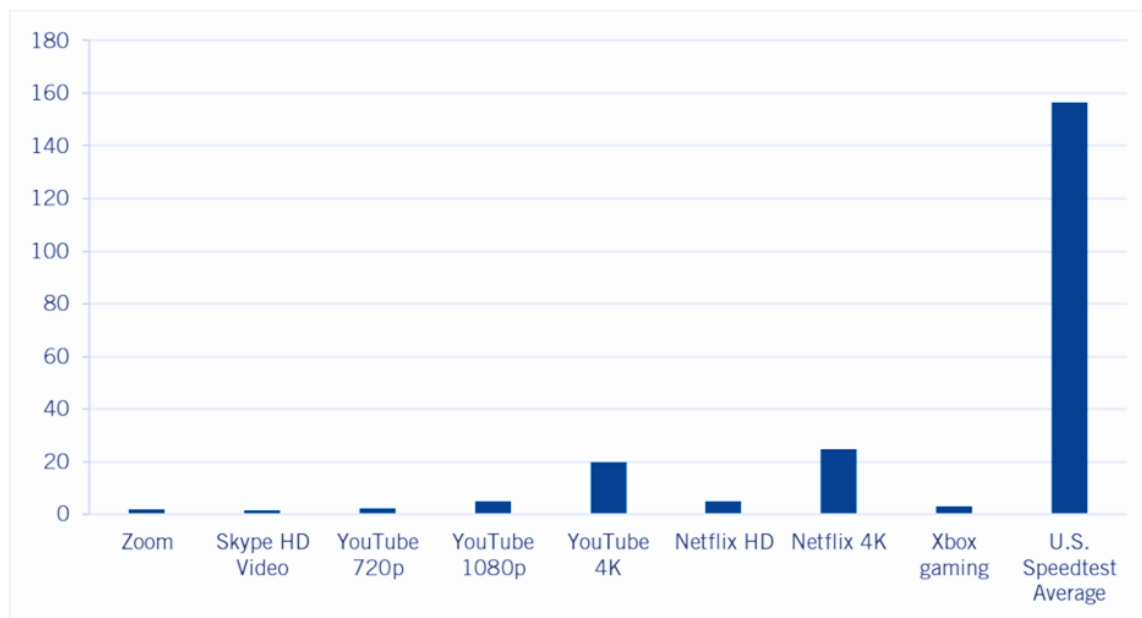
¹⁶ <https://www.fcc.gov/consumers/guides/broadband-speed-guide>

Necessary and Recommended
Bandwidth by Application



Example 2: Digital Republic Suisse: How much bandwidth do I really need?¹⁷ It shows values of SSL as a graphic. It distinguishes between minimal and recommended values, but does not discuss multiple users sharing a link.

¹⁷ Digital Republic Suisse (2022) 'How Much Internet Speed Do I Really Need?: Bandwidth by Application' (2022, January 11). Retrieved from <https://digitalrepublic.ch/en/news/how-much-internet-speed-do-i-really-need>



Example 3: ITIF Speed Needs estimate.¹⁸ It shows values of SSL as a plot, with a supporting narrative.

ITIF takes the position that the speeds being offered in the market today, and which are being advocated by some parties, are in excess of what is needed. We are including this table not in specific support for their position, but to illustrate the representation of SSL information as a plot. The plot does not bring out the points that multiple users may be streaming at the same time, nor the need for headroom. They address these issues in the narrative associated with the plot. They say:

For example, if we assume a household with five individuals all simultaneously watching YouTube in 4K resolution, the household would only need two-thirds of the current average tested speed—a tenth of the gigabit per second advocates call for. A household that can afford five 4K screens can probably afford a decent broadband connection—policymakers may be more concerned about the bandwidth needs of a more modest scenario closer to reality for most Americans: If the same family were willing to watch their video in 1080p high definition, rather than ultra HD 4K, the five streamers would only require 25 Mbps.

Speed	Suitable Activities
0 - 5 Mbps	-Basic web browsing -Checking email -Streaming music on one device
5 - 25 Mbps	-Streaming standard definition video -Online gaming for one player -Video calling for one user

¹⁸ Doug Brake, A. B. (2020). Broadband Myths: Is It a National Imperative to Achieve Ultra-Fast Download Speeds? Information Technology and Innovation Foundation. Retrieved from: <https://itif.org/publications/2020/11/16/broadband-myths-it-national-imperative-achieve-ultra-fast-download-speeds>

25 - 100 Mbps	-Streaming HD video on multiple devices -Online gaming for 1-2 players -Downloading large files
100 - 500 Mbps	-Streaming UHD video on multiple devices -Online gaming for multiple players -Ultra-fast downloads
500+ Mbps	-Simultaneous streaming, online gaming, and downloads on nearly unlimited devices

Example 4: Consumer Reports: Recommended download speeds for different applications. This table makes explicit that they consider the number of simultaneous users, as well as the specific application.¹⁹ This is another example of a tabular presentation of an SSL.

The point of these four examples is to illustrate that information about an SSL can be presented in several ways. Any serious effort to design an SSL should include user testing and surveys to understand how consumers understand and process the presented data. As the literature on actual nutrition labels makes clear, developing a successful ‘nutrition label’ and the associated interpretive data will require cycles of design and evaluation.

Defining the measurement: speed to where? A basic issue about a performance measurement such as speed is that the achievable value depends on the points between which the measurement is defined. In the template label described in the NPRM for wireline providers, there is no discussion of how speed would be measured (e.g., between what points). The definition must be clear about this issue.

It should be well-understood that an ISP that offers a 100 mb/s service is not guaranteeing that a consumer can achieve that speed to any point in the Internet. The Internet is a global system, and any given ISP is responsible for only a small part of it.

The most limited definition is that the ISP is committing only to provide that speed across the access link, the “first hop” from the home to a connection point of the ISP. This definition has the clarity that speed across this link is under the control of the ISP, and thus is a quantifiable metric that is amenable to “truth in advertising” testing. However, if the goal is to help the consumer understand whether the service being offered by the ISP will support a high level of QoE, access link performance is only part of the answer, because the region of the Internet over which the ISP has control is larger than just these links. In the network of the ISP, actual throughput may be affected by network congestion.

Network congestion

While providers of wireline broadband access do typically make specific speed commitments for the access service they provide, they do not make fixed capacity commitments in the rest of the network—to do so would be unrealistic for several reasons. Instead, they provision the links of their network with the goal of providing enough capacity, in aggregate and for most of the time, such that traffic levels do not exceed the provisioned capacity. However, different ISPs may set different provisioning targets, and traffic patterns across the Internet are subject to fluctuations in volume and distribution. Even in well-provisioned networks, there will be periods of congestion, where demand exceeds supply, during which transfers will slow down.

As part of giving the consumer a complete picture, it would be ideal to provide some simple assessment of the extent to which congestion might happen. However, we find no simple answer that is meaningful. First, a congestion event will not typically affect all an ISP’s network, but just specific links. So, some consumers might be affected, while others never notice. Second, episodes of congestion do not

¹⁹ Andy Bergmann, J. K. W. (2021). How Much Internet Speed Do You Need? Consumer Reports. Retrieved from <https://www.consumerreports.org/internet/how-much-internet-speed-do-you-need-a1714131782>

necessarily impair QoE. The relationship between congestion and QoE is complex, and differs from application to application. Third, congestion events are likely to be transient.

For this reason, we propose that the definition of download speed be restricted to the access link—the first segment of the path to the Internet provided by the ISP.²⁰ However, this specification only provides the consumer with part of what they need to know, and further attention to complementary throughput metrics is an important part of performance assessment, and should be part of a larger program of data collection in support of protecting the consumer. Congestion and its consequences are a complex topic, important to understand but difficult to represent in a form that a typical consumer can understand. As well, congestion can have a significant impact on latency as well as throughput. This is an example of a case where the regulator, presumably more knowledgeable and informed than a typical consumer, can gather, process and report metrics that are too complex to include in a simple broadband nutrition label.

We discuss in more detail the issues around congestion in Section 8.

Who should set the SSL?

Once we focus on the need for a sufficient/optimal level for any performance value (such as speed), the next question is who should set that value. In the complex space of human nutrition, the government (the FDA) has taken on that role. We do not want to speculate here on whether it is more complicated and contestable to set DVs for human nutrition or SSLs for Internet applications, but the FCC must pick an approach. If the FCC sets the values, the values can be associated with the label, which will provide the consumer with valuable information at the time of sale. Alternatively, the FCC could invite third parties (e.g., academic researchers, consumer advocates, etc.) to propose SSLs. The drawback of this approach is that the values would not be adjacent to the label at the point of sale, and would be missing at the crucial moment when we hope the consumer will look at the label.

We believe that having information about SSLs at the point of sale is critical to the utility of the nutrition label, and thus the FCC must orchestrate the production of this material. Of course, the FCC can consult with researchers and advocates to develop the best approach. Since the FCC has already proposed a version of an SSL (the Broadband Speed Guide and the associated Household Broadband Guide;²¹ see Example 1), it seems reasonable for them to produce an updated version of this information that could be linked to the broadband nutrition label. This issue was not addressed in the NPRM, but we urge the FCC to take this step.

5. Upstream speeds.

We observe a wide range of assertions for what an SSL for upstream speed might be, ranging from demands for symmetric speeds to modest speeds in the upstream direction. We have not identified any applications today (what would fit into Broadband Definition 1 of the National Academies) that require substantial symmetric speeds.²² We further note that with some technologies (such as fixed wireless links) there is a tradeoff between downstream and upstream capacity. Essentially, from any one base station, there is only so much RF spectrum, and the more that is allocated to upstream, the less there is for downstream. If the allocation must be fixed (e.g., the two numbers must be pre-specified so that “truth in advertising” enforcement can be undertaken), the balance must be carefully considered. (If the balance were dynamic, that outcome might provide better QoE for more application, but the result would be

²⁰ We understand that this specification of how to measure speed differs from the procedures described in FCC DA 18 710, In the Matter of Connect America Fund, WC Docket No. 10-90. We believe that a measurement to verify compliance with requirements that are part of the CAF are different from measurements to inform consumers.

²¹ <https://www.fcc.gov/consumers/guides/household-broadband-guide>

²² Applications such as teleconferencing do have similar upstream and downstream requirements, but the requirements are modest.

harder to explain and harder to regulate. The FCC must consider what the real goal should be. You get what you measure for.)

The most demanding application we find for upstream speed is the real-time streaming of high-quality video (higher quality than simple video teleconferencing), and the context in which this seems to be happening is the streaming of on-line game play, as computer games become a spectator sport. Uploading a video to a site like YouTube need not happen in real time, and thus does not place a specific performance requirement on the upstream service. But real time or live streaming requires enough capacity for the encoded video, and the same considerations apply as for video in the downstream direction. The challenge of matching the service to the application is greater, because upstream services are often slower than downstream services.

A post by Restream illustrates the complexity of matching upstream live streams to service, but makes clear that streaming is practical with today's services with some attention to detail.²³ The simple conclusion from their data is that for real time streaming of HD video at 60 fps, a good SSL is 10 mb/s upstream. Most consumers need a simple answer for it to be meaningful, but sophisticated users with a specific application in mind will be willing to dig much deeper into the details. The nutrition label should target typical users who need information in a simple form.

6. Packet loss rates.

If loss rate is to be reported, there must be a careful consideration of how it is defined, and what an SSL should be. As with other measures, the critical question is over what part of the Internet the loss rate should be measured, and under what conditions.

Due to the way that congestion control is implemented on the Internet (see Section 8), some packet losses are intrinsic to the operation under load. Anyone performing a speed test (which attempts to fully load a path to measure its capacity) should expect to encounter lost packets. Measuring and reporting these losses to the consumer will not be useful. In the past, it was possible to compute a rough relationship between loss rate under load, RTT, and maximum achievable throughput.²⁴ With the evolution of the congestion control algorithm incorporated into the Transmission Control Protocol or TCP, this relationship no longer holds.²⁵ Today there is no meaningful way to map from a measured loss rate under load to an estimate maximum achievable throughput. Our view is that the only useful definition of packet loss is across the access link (the "first hop" from the location of the consumer) when there is no other load on the link. At that point, the measure is an indication of link quality.

Data from the FCC Measuring Broadband America program can be used to get a sense of what loss numbers might be. The "UDP latency test" provides a count of the number of test packets that are lost. In Figure 3, we plot the fraction of lost packets for each of the test points attached to Comcast, one of the ISPs in the MBA program, based on the last validated data sample from 2020. The plots for other ISPs are similar.

Based on the validated data from the FCC, many of the test points encounter very low loss rates, but there are few with a somewhat higher rate. The maximum loss rate across all the MBA test points in this plot was a little above 1%. Based on our analysis of the detailed data from a number of test points, we

²³ Bybyk, Alex, *What is a good upload speed for streaming?*, Restream blog, accessed March 9, 2022. <https://restream.io/blog/what-is-a-good-upload-speed-for-streaming/>. We have no affiliation with Restream. We cite this blog because in our view the information is clear, and illustrates the complexity of the actual SSL.

²⁴ Mathis, Matthew, Jeffrey Semke, Jamshid Mahdavi, and Teunis Ott. "The macroscopic behavior of the TCP congestion avoidance algorithm." *ACM SIGCOMM Computer Communication Review* 27, no. 3 (1997): 67-82.

²⁵ Mathis, Matt, and Jamshid Mahdavi. "Deprecating the TCP macroscopic model." *ACM SIGCOMM Computer Communication Review* 49, no. 5 (2019): 63-68.

conclude that some of the losses may be a transient link impairment specific to that consumer, while data from other links shows a recurring daily pattern that suggests losses due to congestion at peak times. This raises the question of whether losses due to peak period congestion should be included in the specification of loss rate for an access link. The MBA tests measure losses (and latency) to servers on the network, not just the access link, so it is difficult to localize the congestion that is causing the losses.

Measured or nominal?

In defining a basis for a value in the nutrition label, we argue that the published number should be a nominal or target value for that parameter, rather than reflecting actual performance measurements. We think that any “truth in advertising” that tests the actual performance against the nominal value should be separated from the nutrition label. There are many problems with including actual performance measures in the nutrition label, including the question of how often to change the reported value. However, there is no tradition of setting a nominal or target value for packet loss rates as part of the service specification provided by an ISP, so a basis for such a number must be determined. It must be consistent across all providers, and it must allow for some sort of matching “truth in advertising” measurement by third parties. In other words, any description of a nominal or target packet loss rate must be accompanied by a discussion of how measurement can be used to confirm that the ISP is meeting the target.

In the Appendix to this paper, we discuss in more detail our analysis of the MBA loss data, the different ways it could be analyzed, and the resulting implications for some sort of nutrition label.

The MBA data can be used to explore a number of potential specifications, along with a matching measurement protocol. The example nutrition label in Figure 1 uses the term “typical”, but the meaning of that word would have to be defined precisely. If one were to take the *mean* or “average loss rate” across a provider by looking at the measured loss rate of all the customers of that provider in the MBA cohort, that number would be strongly influenced by the fraction of customers suffering from some link impairment or congestion at the moment, which (of course) can change over time.

Another possible specification for loss would be the fraction of users for which the average loss rate is under some threshold. (In this plot, 99.6% of the test points (all but 2) had a loss rate under 1%, and 81% had a loss rate under .1%.) However, as we discuss in the Appendix, this measure is problematic because the exact percentage strongly depends on how many of the high-loss units are excluded from the data. The *median* of the distribution is less influenced by the high-loss units.

A discussion of SSL would talk about the impact of loss numbers that are substantial, but would not focus on comparing essentially similar numbers for reliable link technologies. One problem with the median (based on the MBA data) is that the median loss rate is low enough that the difference we find among providers is probably not material to the resulting QoE. It is important not to trigger an unnecessary “less is better” competition among providers once the rate is low enough.

We conclude that if some nominal or target value for packet loss rate is mandated, that it must be measured across the access link, and that it must reflect the experience that “most” of the users (perhaps 80%) will encounter. However, this must be accompanied by a careful discussion of the SSL for packet loss. See the Appendix for further discussion of loss rates.

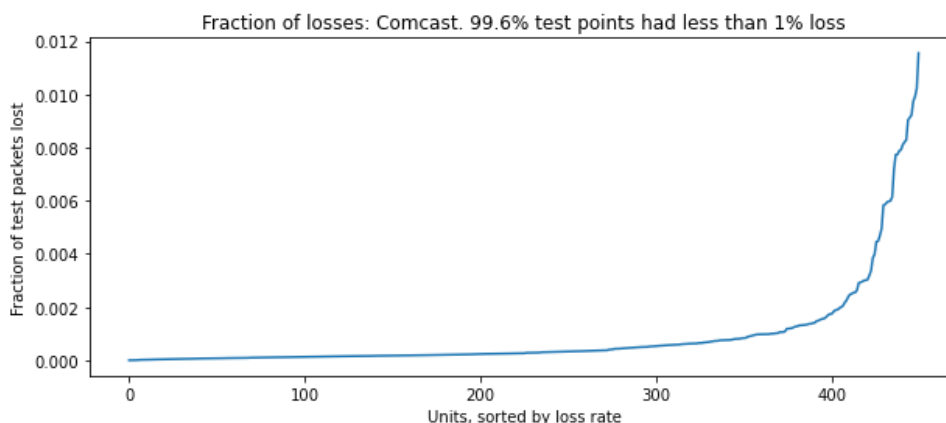


Figure 3: Distribution of loss rates for different MBA test points on the Comcast network. Plot derived from validated data from 11th Fixed MBA Report, released December 31, 2021. The plot suggests that for most users, the loss rate is essentially zero, and for a few users, there might be a minor problem. Plotting the mean of these values to compute some sort of “average loss rate” is nonsense.

7. Latency

The next performance metric mentioned in the FCC NPRM is latency. Latency is the delay between when a packet is sent and when it is received. Latency is sometimes reported one-way from sender to receiver, but also as a round-trip measure (Route Trip Time or RTT), because the typical tools used to measure latency (such as the well-known command `ping`) can only measure round trip times. Any discussion of latency must make clear whether the reported value is one way or round trip.

For many wireline access users today, access speed is no longer the limiting factor with respect to QoE. The importance of latency is getting increased attention.²⁶ The challenges of measuring and reporting latency have the same characteristics as the measure of speed, but if anything, are more complicated, which creates even more tension with the goal of simplicity.

Latency is not being used by ISPs to distinguish between service tiers. The only purposes we can identify for having a measure of latency in the nutrition label is to allow consumers to compare different ISPs, or to allow them to tell if they can use certain latency-sensitive applications with the service being described. Put differently, latency only makes sense as a number to report to consumers if it is accompanied by a discussion of SSLs related to different applications.

We suspect that most consumers have little if any conception of how to interpret a measure of latency (even less understanding than with speed), and thus a reported number with no companion discussion of an SSL is worse than nothing.

A measurement related to latency is jitter, which is the short-term variation in latency that a sequence of packets encounter as they flow from sender to receiver. Jitter is important because, as with latency, some applications are sensitive to jitter. Again, in giving the consumer a simple answer, we must make sure that the answer, once simplified, is still meaningful. We note that there is no mention of jitter (or any measure that captures any aspect of jitter) in the NPRM, and thus again the proposed metrics give consumers only a partial insight into QoE.

²⁶ See, for example, Broadband Internet Technical Advisory Group, *Latency Explained*, 2022, available at <https://www.bitag.org/latency-explained.php>

SSLs for latency

How do latency and jitter relate to QoE? In this case, smaller is better, so the SSL will be some sort of upper bound, not lower bound. But the ideal SSL will depend on the application in question, with some applications essentially insensitive to the typical levels of latency we see on the Internet today, and others critically affected by latency and jitter.

Latency to where? A defining question, as with speed, is latency to where? This question is more complicated than the measure of speed, because several factors contribute to the overall latencies we see on the Internet today. If consumers are to make sense of a latency measure, they need some intuition about actual values and the most common contributing factors. The most important are the following:

- Different access technologies (e.g., FTTH, HFC, DSL) may have different intrinsic latency characteristics.
- Once connections go across a significant geographic area, the speed of light becomes an irreducible limit. The RTT of light across the country (as in signals in a fiber) is almost 50 ms, or a twentieth of a second!²⁷ Typical RTTs we measure across the country as a part of our research are in the range of 75 ms. So across long distances, the Internet today is almost going at the speed of light.
- The third source of latency, which is variable and manifests as jitter, is the delay packets encounter when they reach a congested link and must “wait their turn.” This is the issue of congestion that we mentioned above in the discussion of download speed, and return to in Section 8.

We suspect that consumers have even less of an intuition about latency than about speed, so without some sort of “rough and ready” SSL, consumers may again be drawn into a “smaller is better” marketing strategy (like the “faster is better” over-simplification about speed.) We believe, for the most common applications in use today and for typical values for the first of those contributing factors above (access link latency), most consumers need not concern themselves with latency, except where the technology used by the ISP results in very large access link latencies. For most applications, the typical observed latency across most fixed broadband access technologies is small enough that making it smaller will not materially improve QoE.

Here is a “thought experiment” to illustrate the complexity of the decision-making process that a consumer would have to go through to understand the relevance of a latency number. Imagine that one ISP has an intrinsic RTT across its access technology of about 10 ms, with a few ms. of jitter. (These are the typical values that one of us sees when we measure the latency of our own ISP using the ping command.) Imagine that a competing ISP has an intrinsic RTT of 3 ms. with a few ms of jitter across its access technology. For what applications, if any, will this difference influence QoE? How many readers of this paper are clear on the answer to that question? Unless the consumer has some intuition about an approximate SSL for latency, they will not know how to compare these numbers.

²⁷ It may surprise the reader to learn that the speed of light is a practical issue in the Internet. The Internet sends data over fiber optics, by sending a series of light pulses. These propagate, not surprisingly, at the speed of light. To those who remember any of their introductory science class, the speed of light is 3×10^8 meters per second, which seems like an impossibly fast speed. However, if we translate that into different units, the number is easier to grasp. That number is 300 kilometers per ms. The U.S. is about 4,500 km from east to west coast, (or 9,000 km round trip), so it takes light about 30 ms round trip across the country. Further, light slows down in a fiber! For light to cross the country and back in a fiber optic cable takes around 45 ms.

Latency SSL for a range of applications

As with speed, deriving an SSL must start with the requirements of various applications. Again, as background, we provide a discussion of some relevant applications. Some narrative of this sort, carefully crafted for simplicity, would have to be a part of describing the latency SSL to the consumer.

Teleconferencing: During the pandemic, many consumers have resorted to teleconferencing applications such as Zoom, Skype, WebEx and the like. In real time spoken interaction, excessive latency can degrade the QoE. As early as 1988, the ITU (then the CCITT) did human factors research to determine how much latency could be inserted into a telephone call before the QoE started to degrade.²⁸ At that time, the concern was a returning echo that would distract the speaker. Their answer in that first version of the standard was 150 ms. one-way. In a later version of the standard, they do not measure the effect of echo, but remove all echo and describe the reported satisfaction as a result of the pure delay only.²⁹ We summarize the different tiers of satisfaction they measure as a function of one-way latency in Table 1. The ITU's target application is international telephony, and it is not clear exactly how these numbers would relate to a video teleconference. They are listed below:

One-way latency cutoff for rating	Rating
< 200 ms.	Users very satisfied
< 280 ms.	Users satisfied
< 385 ms.	Some users dissatisfied
< ~500 ms.	Many users dissatisfied
Above ~500 ms.	Nearly all users dissatisfied

Table 1. Levels of user satisfaction with different values of mouth-to-ear delay.

How would the total delay budget for a current teleconference application such as Zoom compare to these target latency bounds. The traffic between participants in most teleconferencing applications does not flow directly from speaker to listener; it goes to a server, where the audio streams from the different speakers are mixed into a single output stream that is then sent to all listeners. Imagine a worst case where two participants on one coast are communicating using a server on the other coast (as might happen if the conference was started by a host on that other coast.) Comparing this case with the latencies above for a cross-country teleconferencing session (75 ms. cross-country and perhaps 10 ms at each end due to the access technology), we see that typical latencies from one participant to the other due to the network would be under 100 ms. The requirements specification for the Zoom app calls for a one-way latency under 150 ms.³⁰

However, these numbers do not equal the total “mouth-to-ear” delay of the application; we need to take into account the delay caused by the audio mixing step in the server and the coding and decoding at the endpoints. We did a quick measurement of the total mouth-to-ear (or microphone-to speaker) latency using Zoom, and found that total latencies, while variable, were around 175 ms. for a nearby server, and around 250 ms. for a far-coast server. Most of the total latency perceived by the participants arises from

²⁸ ITU, CCITT G.114 (11/1988) Mean One-Way Propagation Time, https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-G.114-198811-S!!PDF-E&type=items

²⁹ ITU-T, G.114 One-way transmission time. https://www.itu.int/rec/dologin_pub.asp?lang=e&id=T-REC-G.114-200305-I!!PDF-E&type=items

³⁰ <https://support.zoom.us/hc/en-us/articles/202920719-Accessing-meeting-and-phone-statistics>

the mixing and coding function, not the network latencies. Looking at the ITU table, the total bi-costal latency is in the “satisfied” but not the “very satisfied” region.³¹

The baseline latency of the access network is not the major contributor to the QoE of the typical teleconferencing application. Referring to our thought experiment above, adding or removing a few ms. of latency will not make a material change in the experience. The network factor that disrupts teleconferencing today is the variable contribution to latency (the jitter) that arises if there is congestion in the network and packets queue to wait their turn to be sent. If the queue lengths at the point of congestion are excessive (a problem popularly known as “bufferbloat”) the degradation to the QoE can be substantial. We discuss the problems of congestion and bufferbloat in the next section.

We note that the FCC NPRM on the nutrition label makes no mention of jitter or bufferbloat. These issues, while important, are probably not amenable to presentation in a simple form targeted to typical consumers.

Online performance of music: The most demanding (in terms of latency) Internet application that we can find is the real-time distributed performance of music over the Internet. We have encountered several people who said that during the pandemic, the inability to sing or play instruments together was a major loss for them. The problem they face is that if the different performers hear each other delayed in time, the joint performance falls apart. One developer that targets this application provides the following estimates of desired total one-way latency:³²

- 20 ms. one way: excellent experience, with no impairment.
- 20-30 ms. one way: a good experience, but some latency detectable.
- 30-40 ms. one way: only a fair experience, with the need to explicitly compensate for latency.

These numbers are one way. So, assume our target for an excellent experience is 40 ms. round trip. Our hypothetical example above of the two ISPs had a difference of perhaps 7 ms. round trip. This difference might matter, depending on the other sources of latency between the performers, but even here, the difference in the two access technologies is probably not the most important factor. In a metro area, joint performance of music can probably work if the app is carefully designed (e.g., coding and mixing latencies are minimized); across country the speed of light will prohibit it. The mixing latency of the application must be carefully engineered to be minimal.

For almost all consumers, the difference between our two access technologies (10ms. plus, jitter vs. 3 ms. plus, jitter) is not material. One is not better than the other. On the other hand, an access technology with an intrinsic round-trip latency of 40 ms. would absolutely destroy the QoE of joint music performance. Without some simple SSL, how is the consumer to know?

Online multi-player games: Another class of application that is sensitive to latency and jitter is real-time multi-player games. Highly interactive games (such as first-person shooter games) are highly sensitive to latency. As the latency goes up, the quality of the interaction goes down. As well, this sort of game play is highly sensitive to short episodes of elevated latency (jitter) due to momentary congestion along the path from the player to the game server. A short period of elevated latency, especially if aggravated by bufferbloat (with a duration of perhaps much less than a second) can essentially prevent a player in a battle scenario from defending himself, and cause the death of the player’s character. Gaming is thus sensitive to extremes of jitter, not some measure of the mean.

³¹ The authors note that as users of Zoom, our actual experience with the bi-costal performance is “sometimes unsatisfied.” The issue has to do with the dynamics of turn-taking, when one speaker pauses and another speaker starts.

³² <https://jamkazam.freshdesk.com/support/solutions/articles/66000122532-what-is-latency-why-does-it-matter->. We have no affiliation with JamKazam. We cite their page because we find their explanation of the issue clear and plausible.

While gaming may not seem to be a critical application (though it is a popular one), it is a precursor to future applications that involve highly interactive multi-party interaction in the context of virtual or augmented reality. These speculative applications, which are sometimes described using terms from science fiction such as “metaverse,”³³ will require much attention to control of latency, and are an excellent illustration of what might fit within Broadband Definition 2 of the National Academies (see above).

Technologists fully versed in the concepts and terminology of networking must remember that the typical consumer is not going to master this complexity for the purpose of choosing an Internet provider. Among the latency metrics proposed, we believe that latency of the access link is the most likely to be easily accessible to the largest number of people. Any attempt to define a measure of latency to further points in the network of the ISP will fail the tests of simplicity and meaningfulness. Yet, informed engineers, regulators and application designers must pay attention to the larger issues of latency, congestion and bufferbloat. We will return to this aspect of latency after discussing network congestion.

8. Congestion

In relating network behavior to QoE, the issue of congestion on the Internet is perhaps one of the more complex phenomena that needs to be understood. “Congestion” is the term used to describe the situation where (either as a transient or as a persistent condition), the rate of arriving traffic at a link exceeds the capacity of that link. Any forwarding element in the Internet where congestion might occur (a router, a level 2 switch, etc.) includes a queue or buffer where packets can wait their turn if there is excess traffic. If the congestion event is very short, the buffer might partially fill, and then drain as the congestion abates. However, if the congestion persists, eventually the buffer will fill up, and if the buffer is full when a packet arrives, the router discards the packet. The term congestion is sometimes used to describe the period where the buffer is not empty, or used to describe when a packet is dropped. The definitional nuance is not important—what is important is the phenomenon.

One might assume that congestion is to be avoided if possible. In fact, transient congestion is an essential element of how the Internet operates today. The sending of data over the Internet is controlled (most commonly) by a protocol called Transmission Control Protocol or TCP. TCP is responsible for keeping track of packets sent, resending any packets that are lost, putting them back in order at the receiver, and thus passing on to the receiving application a reliable, loss-free version of what was sent. As well, TCP has another function, which is to determine how fast to send packets. This function is called congestion avoidance or congestion detection.

The Internet does not provide any way for a sender to query the network to find out how fast it can send. There is no global algorithm that computes a real-time measure of available bandwidth between all pairs of senders and receivers on the Internet—such a scheme would be practically impossible to design on a global scale. Instead, a sending TCP must experiment with sending rates to find the fastest acceptable rate. To over-simplify, a typical design works as follows: A sending TCP transmits a certain number of packets and waits for a return indication of whether they were all received (which will typically return in one round-trip). If all of the packets were successfully received, it increases the number of packets it sends in the next round trip, and so on until it discovers that one or more of its packets was lost, which is a signal of buffer overflow and congestion. At that point, it reduces the number of packets it transmits in a round trip, and starts the experimental cycle again. In other words, transient congestion events are a necessary part of the control algorithms that regulate sending rates.

The description above is a great simplification for the congestion avoidance schemes in a modern TCP. The designers of transport protocols such as TCP face a considerable challenge: create a congestion

³³ Stephenson, Neil *Snow Crash*. (1992), New York, Bantam Books, 1993 p.24 ISBN 10: 0553562614 / ISBN 13: 9780553562613

avoidance scheme that works reasonably well across a range of speeds from a 56kb modem (or worse) to a multi-gigabit path, and a range of latencies from a fraction of a ms. (in a data center) to a large fraction of a second (for a trans-oceanic link). There are many variants of congestion avoidance schemes in operation today over the Internet.

Consequences

What are the consequences of congestion, either transient or persistent?

First, the sending TCPs encountering congestion will slow down in an attempt to match their collective sending rates to the capacity of the congested link. This slowdown may or may not have a material impact on QoE.

Second, since packets spend time in the buffer when there is congestion, the latency from sender to receiver goes up. Network engineers typically do not talk about the size of a buffer in terms of how many bytes it can hold, but in terms of how long a packet that arrives when the queue is almost full will wait in the queue until it reaches the head of the line and is sent. Thus, a designer of a router might talk about “a 100 ms. buffer,” which means that the latency across that link might go up by 100 ms. before a packet is dropped and the senders slow their transmission rates. Again, this may or may not have an effect on the QoE of an application, but adding 100 ms. of latency to an application such as teleconferencing can impact the perceived QoE, even if the event is transient.

Third, since there may be many flows that are contributing to the joint congestion of a link, and all of these flows typically share a common buffer at that link, the behavior of one flow can affect the others. Flows across the Internet perform more or less independently so long as they do not encounter congestion, but if they pass through a common point of congestion, their behavior becomes joint. Imagine an implementation of TCP that, while it slows down if it gets a congestion signal, speeds up again more aggressively than flows from another implementation. It is possible for one flow to essentially starve another and capture all the link capacity for itself.³⁴

At the present time, the mechanism that disciplines this behavior is social pressure on TCP developers to comply with well-understood norms of behavior, which have sometimes been called “the TCP-friendly” design rules. But this social mechanism is potentially fragile, and today we see TCPs with different congestion detection schemes interacting with each other in potentially unstable ways. There has been a persistent concern that a developer might release a system that does not respond to a congestion signal, which (if it became popular) could lead to congestion collapse in the Internet.³⁵

The debate around network neutrality seems to be based on an assumption that if ISPs do not implement any traffic management tools, the resulting behavior among different senders will be in some sense equitable, if not exactly equal. In fact, there is no generally accepted definition of what equitable or equal treatment might be. But the Internet is not intrinsically equitable.

Location

Where does congestion typically occur in the Internet? The most common location is the broadband access link that connects the user to the rest of the Internet. As we discussed above in the sections on download and upload speeds, wireline providers usually limit throughput to a specific rate associated with the service tier. It is usually that link that limits throughput, so that is where a flow will encounter congestion.

If congestion occurs at the access link, some technologies help to avoid interaction among users. Schemes such as 4G LTE and the cable system DOCSIS provide each customer with (conceptually) a different buffer, so the traffic behavior of different customers do not normally influence each other. If the

³⁴ Arun, Venkat, Mohammad Alizadeh, and Hari Balakrishnan. "Starvation in end-to-end congestion control." In Proceedings of the ACM SIGCOMM 2022 Conference, pp. 177-192. 2022.

³⁵ Baker, Fred, and Gorry Fairhurst. IETF recommendations regarding active queue management. RFC 7567. 2015.

total capacity of the access technology (the cell tower or the cable capacity) is exceeded, then all active users will perceive a slow-down, but if two customers are each fully utilizing their allocated share, there will be minimal interaction. For this reason, the cross-flow interactions will normally only involve flows from the same customer, as for example when one user in a home is playing a latency-sensitive game and another starts a bulk data transfer.

Another location in the Internet at which congestion may occur more often (or where observers suspect that congestion will occur more often) is the links interconnecting different regions of the network. The location, provisioning and management of these links is a shared responsibility of the two interconnecting parties, rather than a single ISP, and the arrangements are shaped by business considerations.

Since 2016, the Center for Applied Internet Data Analysis (CAIDA) at UCSD and the Computer Science and AI Lab (CSAIL) at MIT have been conducting independent measurements that attempt to assess the degree of congestion on as many of the interconnection links as possible from major U.S. ISPs.³⁶ We observe that the overall level of congestion has changed over the last several years (in general, episodes of congestion at these locations have diminished).

Buffer sizing

All forwarding devices that may encounter congestion on their outgoing paths will have a buffer to deal with the arriving traffic during a congestion event. But how big should that buffer be? That is a critical question, because it determines how much variation in latency (jitter) congestion can cause on that path. If the buffer is 100 ms. in size, the maximum amount of jitter the buffer can add is 100 ms.

There was a rule of thumb, based on the design of congestion avoidance in the implementation of TCP that was in use perhaps 10 years ago, that the buffer should be as large as the typical round trip latency between sender and receiver. So for trans-continental paths, where (as we discussed above), the baseline latency is around 70 ms, a buffer along that path should also be 70 ms. Designers often rounded this number up, and a 100 ms. buffer was a common implementation.

In the core of today's Internet, the buffers in routers are often much smaller. But a 100 ms. buffer at their access link (where congestion is most often expected) is still common and still justified.

It is possible to perform an experiment that gives insight into the actual size of the buffers at each end of a customer access link. One can use the ping tool to measure latency from the home to the other end of the access link. Taking a measurement perhaps every 30 seconds for a minute gives a good picture of baseline latency. Then, while running the recurring ping test, trigger a speed test. A speed test is designed to push the link to its full capacity, so unless the speed test is mis-configured, it will by intention trigger congestion. Figure 4 shows the result running this experiment from one of our own homes.

³⁶ Claffy, K. C., David D. Clark, Steve Bauer, and Amogh Dhamdhere. "Policy challenges in mapping Internet interdomain congestion." *Journal of Information Policy* 10, no. 1 (2020): 1-44.; Dhamdhere, Amogh, David D. Clark, Alexander Gamero-Garrido, Matthew Luckie, Ricky KP Mok, Gautam Akiwate, Kabir Gogia, Vaibhav Bajpai, Alex C. Snoeren, and KC Claffy. "Inferring persistent interdomain congestion." In *Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication*, pp. 1-15. 2018.; Luckie, Matthew, Amogh Dhamdhere, David Clark, Bradley Huffaker, and K. C. Claffy. "Challenges in inferring internet interdomain congestion." In *Proceedings of the 2014 Conference on Internet Measurement Conference*, pp. 15-22. 2014.

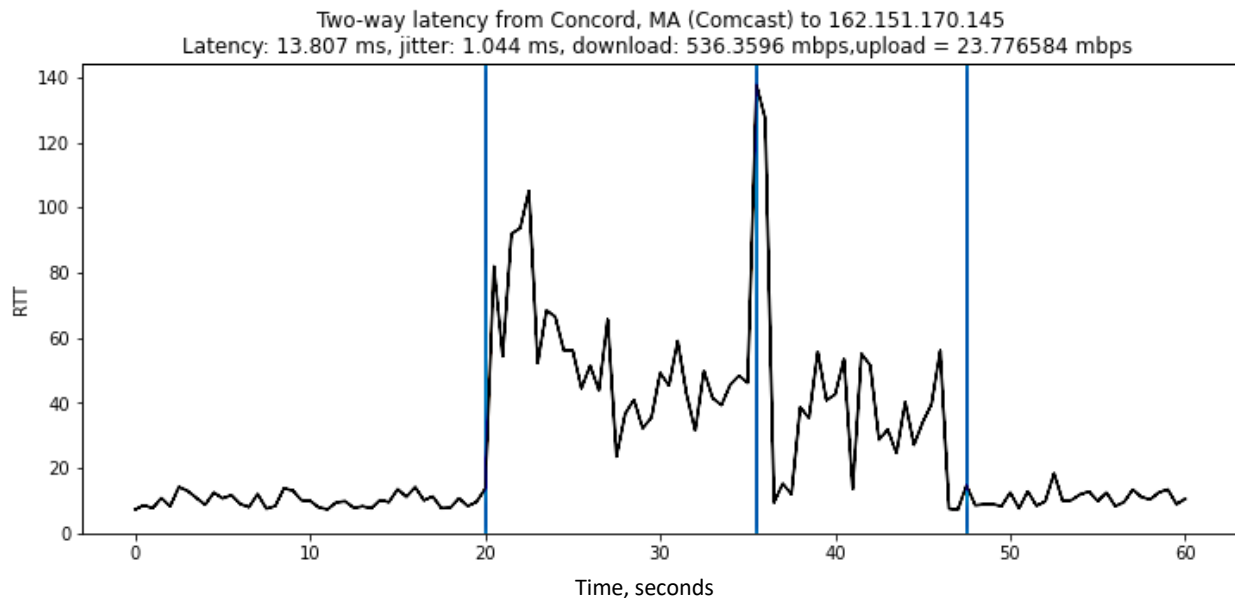


Figure 4: Experiment showing the result of running a speed test across an access link provided by Comcast (running DOCSIS 3.1). The two speed tests (delimited by blue lines) are the download test and the upload test. During the download test, we see variation in latency (jitter) of up to 100 ms. In the upload direction, there is a spike in latency at the start of the test that (in some runs of this experiment) went as high as 250 ms.

The point of Figure 4 is that if someone in a home is running a latency sensitive application such as a game or a teleconference, and at the same time someone does a bulk data transfer (or indeed runs a speed test, which may well be the most congestion-inducing application available to a typical user), the latency sensitive application may see many seconds of jitter that ranges up to 100 ms.

“Bufferbloat”

The previous section illustrated the consequences of congestion over a cable system access link, where the buffer size in the download direction seems to be about 100 ms, and perhaps as much as 250 ms. in the upload direction. That upload buffer seems to be larger than necessary. But different design engineers may have reasons to specify buffers of different sizes.

However, in the Internet today we see locations where the buffers are unjustifiably large. We see buffers that can hold a second or more of data. To our knowledge, there is no reason for congestion buffers of this size, and the cause seems to be design engineers that are ignorant of how congestion avoidance works on the Internet and make the “more is better” assumption. The term “bufferbloat” was coined to describe this phenomenon, and there is an organized movement to inform designers about the issue, and remove buffers that are excessively large.³⁷

Characterizing the prevalence of bufferbloat is difficult. An oversized buffer can be found anywhere on the Internet, but triggering the evidence of that buffer requires running an experiment at a time when the link is congested. Devising a measurement protocol that identifies and quantifies bufferbloat seems challenging.

³⁷ <https://www.bufferbloat.net/projects/> The web site for the bufferbloat project provides a link to a test that a user can perform that is similar to the experiment we plotted above. It is located at <https://www.waveform.com/tools/bufferbloat>.

One approach is a speed test, because they can induce congestion under experimental conditions. The Network Diagnostic Test (NDT) provided by Measurement Lab not only captures the speed of each test, but also two values related to latency: the minimum round-trip time (RTT) and an estimate of the actual RTT at the end of the test. If the test does induce congestion, we would expect the actual RTT to grow during the test as the traffic fills the buffer. By subtracting minimum RTT from final RTT, we can derive an estimate of the buffer size at the point of congestion.

However, this data must be interpreted with caution. The reported final RTT is an estimate (derived from the operating parameters of TCP) and can be highly distorted if there are other problems with the test.

To explore this potential source of evidence about bufferbloat, we use a one month sample of download NDT data from one state in the U.S. (May 2021 in Pennsylvania).³⁸ The sample is large enough to show a reasonable spread of latencies, but by restricting ourselves to a bounded region we can speculate that the range of values are those that a user in that locale might encounter. As a first exploration, we looked at tests where the final estimate of RTT was greater than 1 second. We first looked at data from Comcast, the largest wireline provider in Pennsylvania. We found 9 tests where the final estimate of the RTT was above 1 second and the estimate appeared to be valid.³⁹

The question remains whether these test results are caused by congestion at an oversized buffer. The 9 tests originated from 8 different clients (IP addresses). Some of those only did one test, so there is no baseline testing against which to compare the test with the large latency. However, one client did a large number of tests (23,355), so it is possible to see the large latency tests in the context of the more typical test behavior. See Figure 5a.

³⁸ We further restricted ourselves to tests run using a version of TCP with a congestion avoidance scheme called “cubic,” which we selected because it was more likely to induce a full queue than the other version of TCP (“bbr”) which is now used in many of the NDT speed tests.

³⁹ The final estimate of RTT reported by NDT must be used cautiously. We initially found 286 tests out of 12,403 from 71 different clients that had a final estimate of RTT above 1 second, a tiny fraction of the total tests. (For comparison, the number of tests with a final RTT above 250 ms. was 1,600.) Next, manual inspection of those tests revealed that almost all of them had some other indication of an impaired test: very high loss rates (which can lead to an invalid estimate of final RTT), a test duration greatly in excess of the normal 10 second target for NDT, or very low measured bandwidth, which is another signal of a test that is not a valid measure of the link capacity. We picked a slightly arbitrary set of constraints to filter these measurements: the test duration had to be < 20 seconds (twice the normal NDT test duration), the loss rate had to be less than 2% (a higher value might still represent a valid speed test, but we do not want to rely on the RTT estimator in this case), and an achieved throughput > 10 mbps. For Comcast, essentially all of their customers have a download speed tier well above this limit, so a test that could not run above 10 mbps is flawed in some way. The nature of the flaw that might lead to the poor performance and odd statistics is interesting in its own right, as we have explored in previous work, but with this focus on finding bufferbloat, we reject these tests as an unreliable indicator. When we restrict ourselves to the tests that pass these selection criteria, the set of 286 tests dropped to 9.

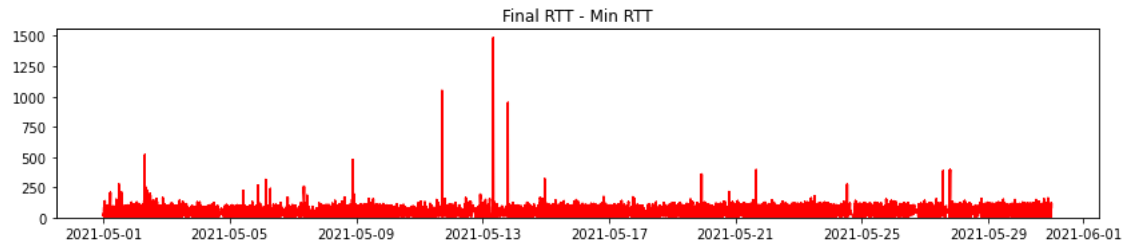


Figure 5a: Plot of (final RTT – min RTT), which is an estimate of the size of the queue formed at the bottleneck along the path from an M-Lab client in Pennsylvania to an M-Lab server in Dallas, TX. The client ran 23,355 measurements during the month we analyze. The typical estimated queue size is usually less than 100 ms, which is consistent with a reasonable buffer size that might be found in the Internet. However, there are a few tests that show a much larger final buffer size, up to almost 1.5 seconds.

Figure 5a shows that almost all the tests resulted in an estimate of buffer size that is less than about 100 ms, but there are a few tests with a much larger estimate. We can only speculate as to how this result might happen, since an end-to-end speed test cannot isolate the location of the congestion. However, here is one hypothesis as to how these results might arise. This flow may pass through two different potential points of congestion. The one that is normally the bottleneck is properly sized at around 100 ms. The other is much larger, but is not normally the bottleneck. However, if some other traffic also transits that buffer during the test, congestion could arise there from the aggregate of the traffic, and in this case this buffer, with its much larger size, becomes the overall limit in the speed test.

The results we see here are not distinctive to Comcast. When we do a similar query for Verizon, the other major wireline provider in Pennsylvania, we find a similar number of high final RTT estimates: when we exclude tests where we consider the final RTT estimate to be unreliable, we end up with 8 measurements. Figure 5b shows a plot from a series of tests performed by a client attached to the Verizon network.

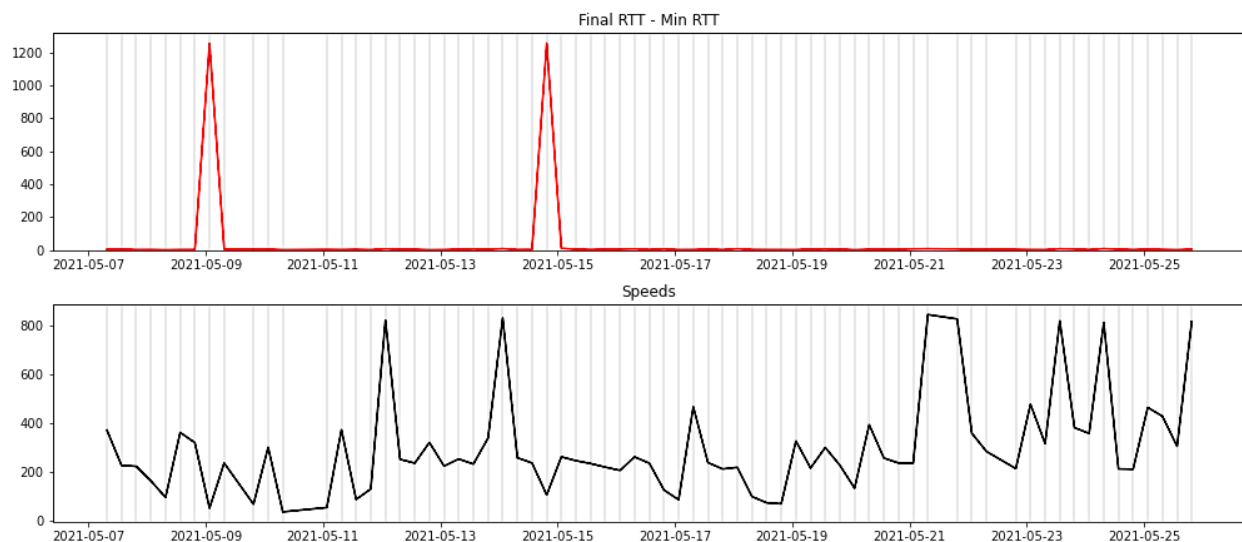


Figure 5b: Plot of estimated final buffer size and speeds for a series of tests (indicated by the light gray lines) from a customer of Verizon during much of the month analyzed. In this set of tests, the typical final buffer size is close to zero, but there are two events where it exceeds a second. In this case the speeds of the tests are plotted. The measured speeds are highly variable, which suggests

that there is some issue with this client. The speeds associated with the tests with high final latency are among the lower values, but not distinctly low. Final buffer size is not necessarily correlated with low measured speed.

In terms of user QoE, we must ask if a potentially disruptive event (1 second RTT) actually is worth worrying about if it only happens a few times a month. What is needed is a more organized measurement campaign to see if there are regions of the network where the phenomenon is much more common. From the perspective of a consumer-facing summary, it is hard to see how to reduce the information in this plot into a single meaningful number. This illustrates our point that the regulator must bring a more sophisticated understanding of the underlying phenomena that influence QoE than can be captured in a simple “nutrition label.”

Sudden increases in latency can be particularly disruptive to Web browsing because to load a web page requires many separate transfers, each of which can be delayed by a bufferbloat episode. In Figure 2, we plotted the median page load time vs. access speed tier using data from the FCC MBA program. That plot shows the median, but does not give any sense of how much variation there was from experiment to experiment. Figure 6 plots the actual distribution of load times for the experiments—we plot data only for Comcast, to avoid an impenetrable plot. The plots for the other providers were not materially different. The box in the plot shows that the 25th to 75th percentile of the results are tightly clustered around the median, but isolated page downloads could take as much as 25 or 30 seconds. While this could be due to transient load on the server, we speculate that this is the result of a bloated buffer delaying many successive downloads that make up the web page. Increasing the speed tier does not seem to improve these slow load times, just as it does not improve the median.

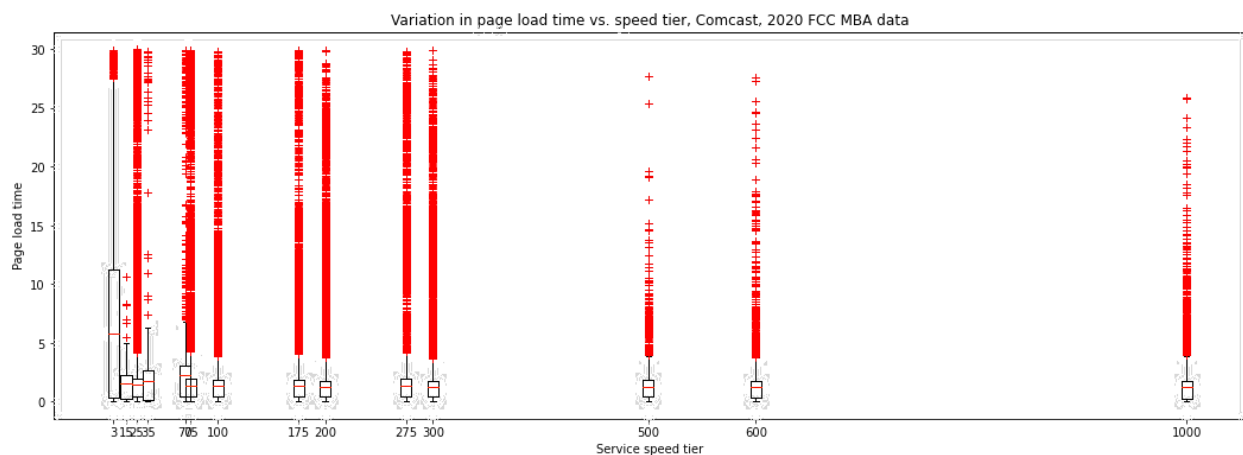


Figure 6: Distribution of page load times as a function of speed tier, 2020 FCC MBA webload data, for Comcast. While the 25th to 75th percentiles (indicated by the box) are tightly clustered for the higher speed tiers, the outliers (red crosses) show that individual page load times can be as high as 30 seconds.

Measuring latency and jitter

The issues that surround latency are more complicated than those related to capacity or throughput limits. A speed specification of “100 mb/s” is easy for both consumers and regulators to understand. The only residual issue is “truth in advertising.” Latency (and its short-term variation—jitter) has more nuance and is harder to reduce to a simple number that a consumer could understand. At the same time, issues surrounding latency can have a material effect on QoE for some applications.

We identify two possible specifications of latency that could be provided to consumers. One is latency when the link is not loaded, which would provide a baseline number for the access technology in use. However, the no-load latency of a modern wireline access link is not normally a material contributor to the effects we discuss here and has little relationship to QoE. The other possible specification is latency while the link is under load—e.g., during a speed test. This latter measurement would reveal the extent to which the access technology being used by the ISP suffers from bufferbloat. The challenge here is to define a measurement method with sufficient clarity and specifics that comparison is useful. At a minimum, the specification must define:

- Latency to where? To the other end of the access link? To some point further into the access network? How can this be defined in a consistent and repeatable way?
- Measurement method. The simplistic use of a tool like `ping` can fail to reveal how actual user traffic is being treated. The result of running `ping` will depend on the exact treatment that the probe packet receives at the far end of the access link.
- Inducing load. Is the latency measurement a repeated measurement under normal operating conditions, or during a test (e.g., a speed test) expected to induce congestion, as in Figure 4? If so, to what extent would the exact behavior of the speed test need to be standardized? Different speed test tools, using different methods, may induce congestion to different degrees.
- How to aggregate measurements taken across many access links into an overall score for an ISP. The degree of buffering (or bufferbloat) in a given ISP may vary depending on the exact equipment in use in different parts of their network.

We see no obvious way to map latency and congestion measures into any sort of overall “figure of merit” for different ISPs that would be simple enough for consumers to understand.⁴⁰ Such an effort would require significant community debate as to how to compute any such simple metric, and would require that a measurement project like this be funded to make a transition from a research effort to a sustainable contribution to consumer education. We do not believe that at the present time there is a way to derive and present a simple measure that captures the level of congestion in a network and the resulting impact on QoE in a way that would make sense to consumers. Latency is an area where a “broadband nutrition label” may not provide much useful insight at the consumer level, but where regulators and the experts they engage must develop a sound understanding of the relevant issues, and use their authority and encouragement to push the Internet in the right direction.

9. Making things better

There are a number of efforts under way to improve the situation with respect to latency and jitter. The first is simply an awareness campaign, targeting designers of network gear, to persuade them to stop putting unnecessarily big buffers in their equipment. The other two are technical advances that are intended to control jitter in the context of “normal” congestion events.

Active queue management

Active queue management, or AQM, is an idea that has been under study for many years, since the first proposal for AQM (Random Early Drop or RED) was proposed in 1993.⁴¹ The general idea of AQM is as follows: a router can detect when a buffer is filling up. Rather than waiting until the buffer is

⁴⁰ The BITAG report “Latency Explained” calls for the reporting of the 99th percentile of latency across a number of tests. We are sympathetic to that proposal—reporting a mean or median will not be useful, since the distribution of latencies cannot be expected to be normal or Gaussian. However, that proposal would have to be fleshed out with answers to our points above.

⁴¹ Floyd, Sally; Jacobson, Van (August 1993). “Random Early Detection (RED) gateways for Congestion Avoidance”. *IEEE/ACM Transactions on Networking*. 1 (4): 397–413.

completely full to drop a packet, start dropping probabilistically as the queue in the buffer grows. This has several benefits: senders start to adjust their sending rate sooner so the queue (on average) is shorter, so the buffer can hold a burst of packets, and there is less coupling between the different flows because the drop signals are generated probabilistically rather than when the buffer is full. The difficulty with AQM is to find a scheme that works across the wide range of conditions (speeds, locations in the Internet, etc.) and does not require manual tuning by an operator that may not fully understand what the tuning parameters are. Modern proposals for AQM are self-tuning and are mature enough to deploy.

Research concerning AQM has continued apace, and a number of proposals that show great promise now exist. Further, there is a key insight that suggests the utility of AQM. Since flows normally trigger congestion at the ends of a broadband access link, if a suitable AQM scheme is installed at the two end-points of the link, the jitter associated with buffering there can be mitigated, without the scheme needing the generality to work at all locations in the Internet. The Internet Engineering Task Force made the deployment of some AQM scheme a Best Current Practice in 2015.⁴² This RFC outlines many of the issues we mention here, including reducing interaction among flows, reducing average length of queues, and the challenge of dealing with non-responsive flows. While there is still resistance to the idea of AQM among some network operators and consumer advocates (perhaps due, in part, to a misunderstanding of how it works) most of those developers working on reduction of jitter consider it an important mechanism to deploy. The details of the schemes do not matter in this context—the important conclusion is that deployment should be encouraged, not hindered by regulation or other barriers.

Figure 4 illustrated the jitter that can occur over a residential access link under load (in that case, a cable access link running DOCSIS 3.1). The home from which we did that experiment used a home router that supported a version of AQM.⁴³ To compare the jitter with and without AQM, we turned on AQM in the upstream direction only and repeated the experiment described in Figure 4. Figure 7 shows no jitter during the upload portion of the test. Of course, one experiment is not a sufficient basis to evaluate a scheme (although this particular result was consistent across many trials of the experiment). The point of this illustration is to suggest that AQM can make a material difference in controlling jitter.

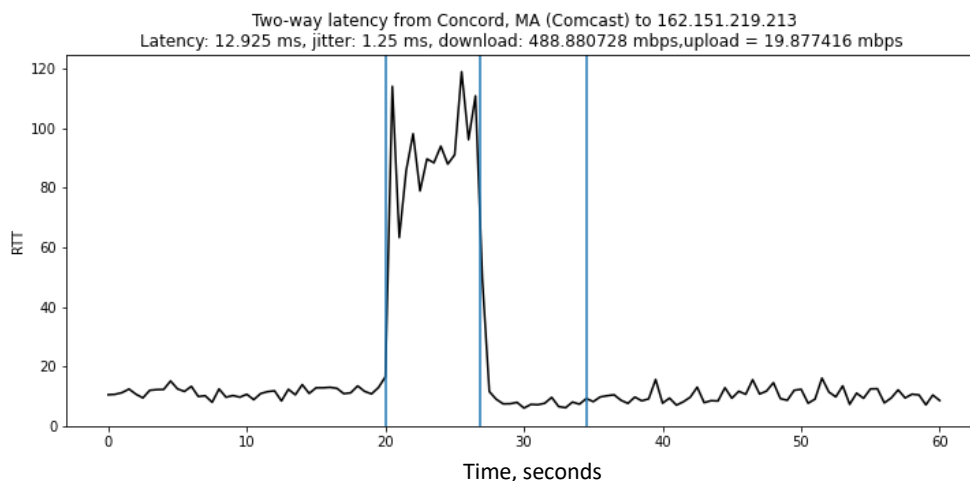


Figure 7: The results of running the experiment described in Figure 4 with AQM enabled in the upstream direction (the second of the two test intervals). Note the absence of jitter.

⁴² Baker and Fairhurst. RFC 7567.

⁴³ The router was an Ubiquiti EdgeRouter X v1.10.7. The variant of AQM supported by this device is FQ-CODEL (Fair Queuing with Controlled Delay) + HTB (Hierarchical Token Bucket)

Multiple queues

Another approach to reducing congestion-generated jitter is based on the observation that many applications currently in use on the Internet do not need to run “as fast as possible.” They do not need the capability of TCP to constantly probe the network to see what the current available capacity is. They deal with congestion in a different way. Consider a typical video streaming application such as YouTube. When a viewer starts a YouTube stream, the app does not download the whole video at once. That would be quite wasteful of network and server capacity, since viewers often abandon a video before it ends. Instead, chunks of the video are fetched a little in advance of the time they are needed. A scheme might, for example, fetch a 5 second chunk a few seconds before it is needed. The sender need not send that 5 second chunk “as fast as possible”, but might pace out the packets at a rate perhaps twice the playout speed of the video. Thus, if the video plays out at 4 mbps, the server would pace the next chunk to a steady flow at 8 mbps. The receiver then looks to see how long it takes the chunk to arrive. If it arrives in 2.5 seconds, there is adequate capacity. If it takes longer than that, the receiver would request that the next chunk be sent at a lower resolution, with a lower playout rate. In this way, the streaming video service can adapt to changing network conditions without needing to build a queue in a buffer. Similarly, a teleconferencing application does not send “as fast as possible,” but at the rate of the encoded audio and video.

This kind of rate adaptation depends on application-level knowledge about the necessary traffic load. TCP was designed to support any application, and thus has a very general (and aggressive) approach to capturing capacity. If the application itself takes on the role of congestion avoidance, the TCP mechanism (even if the application chooses to use TCP) never comes into play. This class of application is called “non queue building,” while an application such as bulk data transfer is “queue building.”

The IETF is developing a proposal called Low Latency, Low Loss, Scalable Throughput (L4S) Internet Service,⁴⁴ which specifies that a router should provide two buffers or queues at the point of congestion—a traditional long queue for the queue building applications and a very short queue for the non-queue building applications. In this way, flows such as interactive teleconferencing and games never encounter the queues caused by bulk data transfers. The goal is that non-queue-building applications should achieve a jitter target of less than 2 ms. This reduction in jitter would bring a significant improvement to jitter-sensitive applications such as teleconferencing, compared to the 50 ms or more that they encounter today.

Again, from a regulatory perspective, the technical details of the scheme are not important. What is noteworthy is that this scheme implies allocation of traffic to different queues, which could run afoul of pressure for network neutrality (if defined in certain ways). The current proposal is that the ISP should not make the decision about how to queue a packet, but it should obey the marking in the packet that is set by the application. In other words, the application, not the ISP, has control over how the traffic is treated. In our view, this sort of traffic management is benign, and would bring a material improvement to the QoE of latency-sensitive applications. Regulation should encourage, not discourage, this sort of traffic management.

10. CONCLUSION

Broadband "nutrition" labels

A broadband nutrition label with no hints about how much is enough will just perpetuate marketing based on the flawed intuition that “more is better” and the potential for profit-motivated upselling that might lead to unnecessary expenditures on the part of consumers—the exact opposite of the label's stated

⁴⁴ Briscoe, Robert, et al, *Low Latency, Low Loss, Scalable Throughput (L4S) Internet Service: Architecture*, IETF Internet Draft, July 2022. This document, now in its 18th revision, represents an area of ongoing, active development by the IETF.

goal. We believe that information about appropriate SSLs should be generated and provided as part of the label itself, because this information must be made available at the point of sale. Otherwise, the data in the label will be floating freely without context that is absolutely necessary to making an informed decision. If these performance metrics are to fulfill their intended purpose, the FCC must orchestrate the creation of this material, must ensure that it is independently verifiable, and guarantee that it is available at the point of sale.

No performance metric should be considered for inclusion in the broadband nutrition label unless the proposal includes a description of the way in which the SSL has been developed to provide the metric. Methods need to be standardized across providers, otherwise there will be no empirical basis for comparison. Any proposal for a performance metric must be vetted for simplicity, meaningfulness, comprehensibility,⁴⁵ and delineation of sound methods for independent verification. The test method must be carefully specified, answering such questions as “measurement to where?”

We have presented specific definitions for the performance metrics in a possible label, along with advice about how an SSL might be crafted. We believe that these achieve a pragmatic balance that reflects the goals of simplicity, meaningfulness and independent verification. However, we stress that these measures do not by any means address all the concerns that consumers may legitimately have about whether the service they are contemplating will allow them to achieve good QoE for them. To be genuinely helpful, the regulators and the expert they engage must have a technically sound understanding of the relevant issues, as a part of consumer protection.

We stress that to allow for the emergence of future applications, the discussion of SSL should consider the Broadband Definition 2 from the National Academies, that broadband services should provide sufficient performance—and wide enough penetration of services reaching that performance level—to encourage the development of new applications

Speed—is more better?

When one considers the *actual* needs of a broad range of consumers, it appears that many are buying into higher-speed, higher-cost tiers unnecessarily. What propels them to do so? This is an open question, but we do have some hypotheses.

One motive may be the same one that prompts some to buy high-performance cars.⁴⁶ But a more relevant reason may relate to latency and jitter. A high value of “latency under load” will cause interactions among applications (e.g., a bulk data transfer and a real-time gaming application) that can disrupt the QoE of the real-time application. We suspect that one reason why consumers are motivated to purchase service tiers with speeds well in excess of the nominal SSL is to try to mitigate congestion and bufferbloat problems. Adding excess bandwidth may actually help to some extent (by reducing the probability that a congestion event will occur). Buying more bandwidth is the only remedy available to users today if there is a problem with their service. They cannot buy a low-jitter service. More speed may be a poor (and costly) remedy, but it is the only option. We think that encouraging a concerted effort by ISPs (and others) to deal with issues of latency and jitter might allow users to be content with a slower service tier. Of course, we note that if this is true it creates a perverse incentive for the ISPs, but we argue that this is the direction in which the Internet should go.

Independent verification

One of our criteria for any metric is that the actual network performance relative to the metric be verifiable by an independent third party.

With respect to upstream and downstream speeds, there are several third-party tools that consumers use today to measure access speeds. The issue with some of those tools is that they are inaccurate under

⁴⁵ By definition, all proposers should plan to include consumer research results, from primary studies they have conducted in their own markets, to show that these requirements have been met.

⁴⁶ Given that this is not a psychology lecture, we shall say no more.

certain circumstances. Speed testing can be conducted many ways - and they are not all equal in thoroughness or accuracy. Which ones are best, and for which uses? *How* a test is conducted, and the ways in which the results are analyzed, can have a tremendous impact on what is reported. We have seen many cases where the methodology and/or data analysis were implemented, often unintentionally, in ways that produced vastly distorted reporting.⁴⁷ In short, if a nutrition label is going to bring increased attention to download and upload speeds, regulators need to consider the problem of poorly calibrated testing programs.

With respect to latency and loss rate across the access link, a sophisticated user today can make a rough measurement of these parameters by (carefully) using a latency measurement tool like ping. One of the reasons that we do not propose measures of latency and loss rate that go further into the network is that if running repeated ping tests across the Internet became a national pastime, the network might suffer from processing all the measurement traffic. However, providing instructions as to how to use ping to measure an access link might benefit some consumers while protecting the network from compulsive testing.

Measure it or fix it?

There is an expression sometimes heard in the security community: “admiring the problem.” A focus on measurement is important: if done properly, it can make us aware of problems; it can help us size them, and identify their precursors and causes. It can help us to redress them, and quickly. It can help us plan, by allowing us to chart trends over time. It can also function as an ongoing outcome study, that helps us examine the impact of any change on QoE. Yet, it is a worthwhile undertaking only if it is done well, and only if it is used to make things better. Measurement is an essential foundation for continuous improvement, but it is not a proxy. The responsibility putting effective measurement programs into place, and for putting their results to use in creating a better future, is a human problem. It is ours, and ours alone.

Funding acknowledgement

Funding for this work came from the National Science Foundation (NSF) grant OAC-2131987, the MIT CSAIL Corporate Alliance Program, and an unrestricted gift from Comcast. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of any of the sponsors.

⁴⁷ Clark, David D. and Wedeman, Sara, Measurement, Meaning and Purpose: Exploring the M-Lab NDT Dataset (August 2, 2021). TPRC49: The 49th Research Conference on Communication, Information and Internet Policy, Available at SSRN: <https://ssrn.com/abstract=3898339> or <http://dx.doi.org/10.2139/ssrn.3898339>

Appendix: A deeper exploration of MBA packet loss data.

The discussion of packet loss in Section 6 drew on data collected by the FCC as part of the Measuring Broadband America (MBA) program. The MBA program uses hardware devices located in the homes of volunteers that perform a variety of tests of their broadband access service, including download and upload speeds, web page load time, a simulation of video streaming, and latency measurements to nearby on-net and off-net servers. The latency measure can also be used to measure packet loss, and this was the data we used.

The measurement campaign is structured as follows. Each hour, the hardware devices send up to 2500 probe packets to the server being used, each of which elicits a response (this is the test that the ping command does). The device records the delay from sending the probe packets and getting the responses, and reports the minimum, maximum, mean and standard deviation of these latency measurements. It also reports the number of probes for which there is no response, which can be used as a measure of packet loss, whether in the outgoing or returning direction.

Since each device starts a new measurement campaign each hour, in a 30-day month there will be as many as 720 campaigns from each device to each server being used (normally 2—one on-net and one off-net).

Our initial starting point was to look at the overall loss rate from each measurement device, to see what the variation in loss rates among the different devices. We plotted the data for each ISP in the MBA program, to see if there was significant variation among ISPs.

We first looked at the so-called “validated” FCC data for September-October 2020, the last month for which they have released validated data. The FCC documents the rules for validation;⁴⁸ it involves removing units that were active for less than 5 days, removing certain days when some anomaly may have been detected, not plotting data for ISPs that had less than 45 active units, and rejecting data from certain servers (targets). For the latency test in particular, they further remove the result for any hour where the loss rate exceeded 10%. The report states:

Such a high level of loss would render a connection unusable and is considered an anomalous event.

Of course, if the purpose of gathering data is to observe the frequency of anomalous events, this step throws out all the interesting data. The issue is whether the anomalous event related to the test itself (a problem with a server, perhaps) or an event related to the network under test.

⁴⁸ Federal Communications Commission, Measuring Broadband America: Validated Data Cleansing, Eleventh Report, available at <https://data.fcc.gov/download/measuring-broadband-america/2021/data-cleansing-sept2020.pdf>

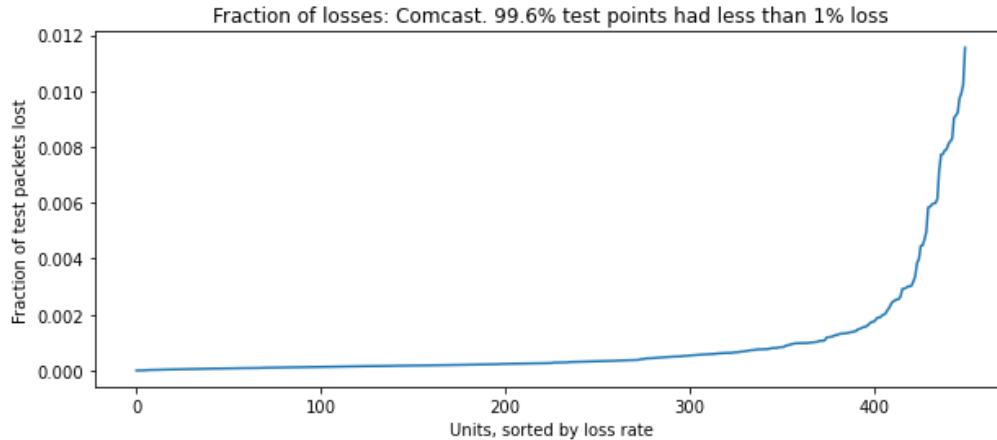


Figure A1: Distribution of loss rates for different MBA test points on the Comcast network. Plot derived from validated data from 11th Fixed MBA Report, released December 31, 2021. The plot suggests that for most users, the loss rate is essentially zero, and for a few users, there might be a minor problem. Plotting the mean of these values to compute some sort of “average loss rate” is nonsense.

We reproduce here Figure 3 from this paper, which shows the loss rate for the different units in the Comcast network that were included in the validated data.

We looked at the data from the specific units that had high loss rates to see if we could infer the cause of the losses. Figure A2 shows the losses recorded from one such unit over the month. One can see a recurring daily pattern (the numbers at the bottom are the hours in the month, so a daily pattern will show recurring episodes that are offset by 24). This plot suggests that the cause of the losses is not a “noisy link,” but peak hour congestion. Since the loss rates to all servers are affected, the congestion was probably occurring near the location of the measurement unit.

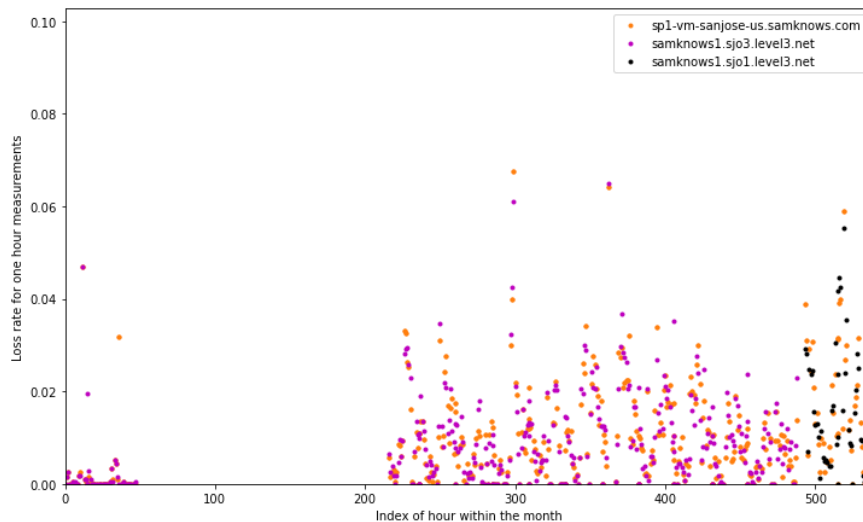


Figure A2: Loss rates over the validated month for one unit (FCC number 40730089). The mean loss rate was 0.011. Note the recurring daily pattern (increased losses every 24 hours).

We were concerned that the somewhat arbitrary removal of the data for any hour for which the loss rate was over 10% might have masked some important examples of anomalous behavior. To explore this option, we downloaded the raw data the FCC provides, and developed our own approach to data

cleansing.⁴⁹ We first rejected data to any server that was unresponsive (of which there were several), and then rejected data from any unit where the average loss rate over the month was greater than 20%. That is, we cleaned the data by rejecting units as well as measurement hours. Then we included all hours from the remaining units where the loss rate for that hour was less than 60%. The 60% is perhaps arbitrary, but this rule threw out only a few hours that had very high loss rates (perhaps the server was down for that hour), and the results were not very sensitive to the exact value. The results from this approach to data cleaning are shown (again, for Comcast—the other ISPs were not materially different) in Figure A3. Comparing this to Figure A1, our approach used data from more test units, and found units with somewhat higher loss rates. The maximum loss rate in this plot is around 4%, as opposed to 1% in Figure A1.

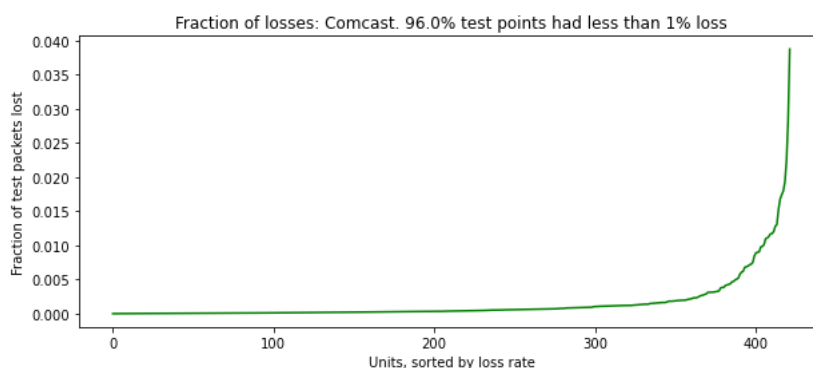


Figure A3: Distribution of loss rates for MBA measurement units in the Comcast network, using our data cleansing approach.

As a way to compare these two results, we plot in Figure A3 the results of the validated data, our cleansed data, and the raw data. The plot clearly shows that the raw data includes units that had a 100% loss rate for the month, which reflects a failure of the test protocol. Both our rules and the FCC rules removed these servers from the valid data. Our cleaning approach (the green line) used data from more units than the FCC validated data (the blue line).

However, for most of the units the results of the three data sets are visually indistinguishable. That is, including hours with a loss rate over 10% made no substantial difference for most of the units.

⁴⁹ While the FCC normally posts the raw data for the validated month once the validate result are released, they seem to have failed to do so here. We were forced to use the raw data from August 2020, the previous month. It is possible this causes a minor inconsistency in the comparisons, but the data for the two months looks very similar.

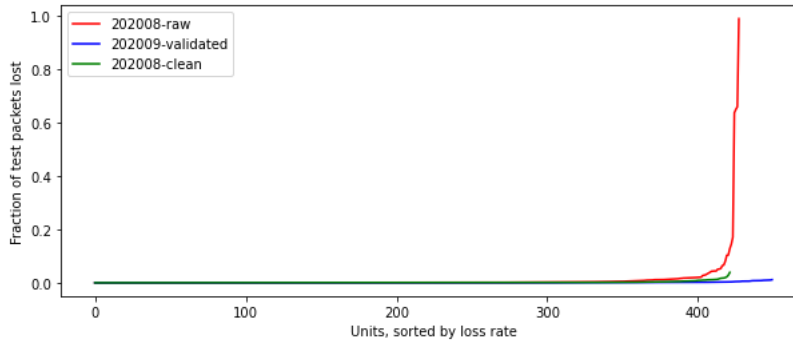


Figure A4: Plots of raw, FCC validated, and our cleaning, for units in the Comcast network. The curves are visually indistinguishable for most of the units, but the plots reveal slight differences in the high-loss units between the FCC and our cleaning. The plot of the raw data, showing 100% loss rates for the month, makes clear that many of the tests are invalid.

Because the FCC nutrition label proposed to list loss rates for different ISPs, we wanted to see how the distribution of overall (monthly) loss rates compared for the different providers in the dataset. Figure A5 shows the distribution of losses for each ISP, plotted as a violin plot. The outliers correspond to the very right side of Figure A4—the few units that had high loss rates.

Verizon DSL had the distribution of losses with the highest values, which is not surprising, as DSL links can suffer from a variety of problems. As an alternative to plotting the data by provider, in Figure A6 we plot the distribution of loss rates with units attached to different access network technologies. Again, not surprisingly, DSL had the highest loss rates, fiber the lowest, and cable somewhat higher than fiber but clearly lower than DSL.

To see the degree to which units with the same technology had different loss rates in different providers, in Figures A7-A9 we plot, for each technology, the loss rates for the different providers of that technology. In a few cases, one provider had a distinctively different profile. Again, some of these higher loss rates were due to daily congestion somewhere on the path from the unit to the servers being used. It is a matter of policy whether this source of packet loss should be folded in to a loss measure, since that would mix losses due to some intrinsic link characteristic, losses due to some transient event, and losses under load.

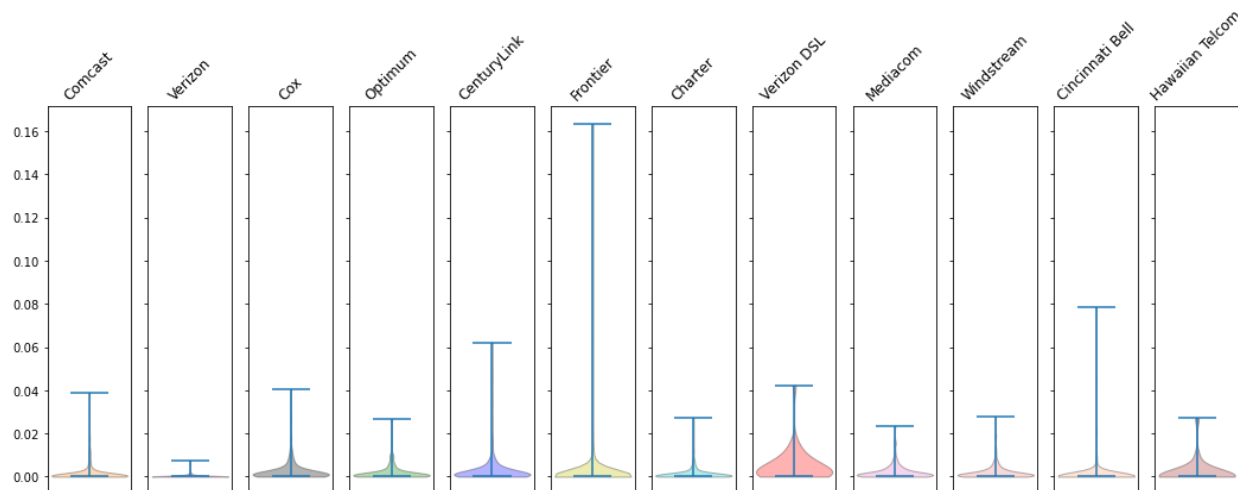


Figure A5: Distribution of overall (monthly average) loss rates for the units in each of the plotted providers, displayed as a violin plot. Verizon DSL shows a higher loss rate compared to the others. Verizon FIOS shows a very low loss rate.

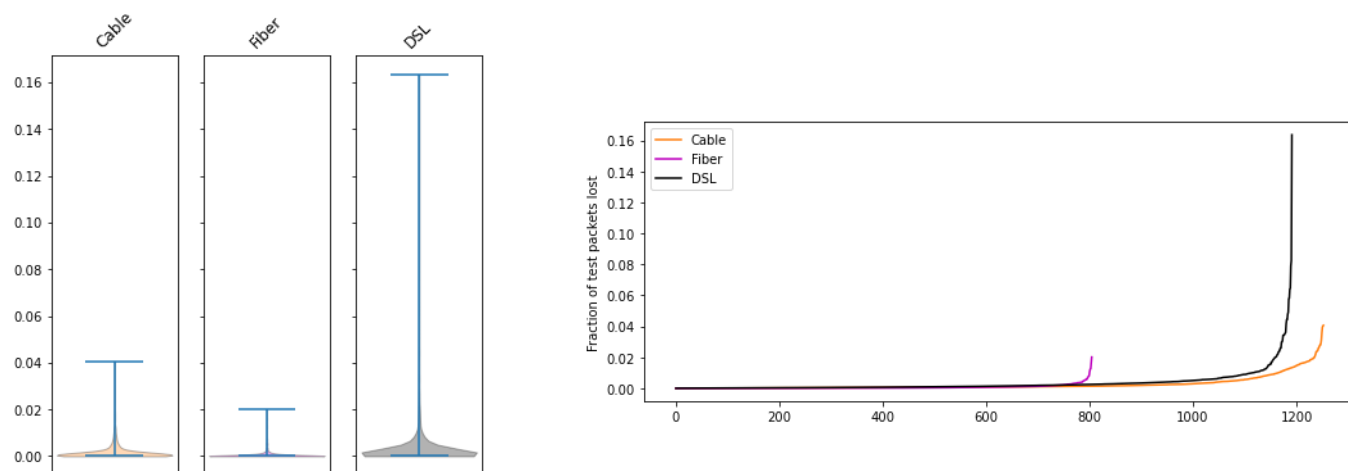


Figure A6: Distribution of losses, by the network technology to which the unit is attached. Plotted both as violin plot and as a probability distribution.

If one were to pick as an overall measure of merit the percent of units that had a loss rate under 1%, this would trigger a necessary debate as to how many of the marginal units to include. The FCC cleaning process rejected more of the marginal units, which resulted (for Comcast) in 99.6% of the units having a loss rate less than 1%. Our cleaning, because it included more high-loss units, resulted in 96.0% of the units having less than a 1% loss rate. This sort of difference is problematic, since it puts a high premium on defining exactly what unit tests should be excluded, on what basis. That argument has nothing to do with the experience of a typical user on the network.

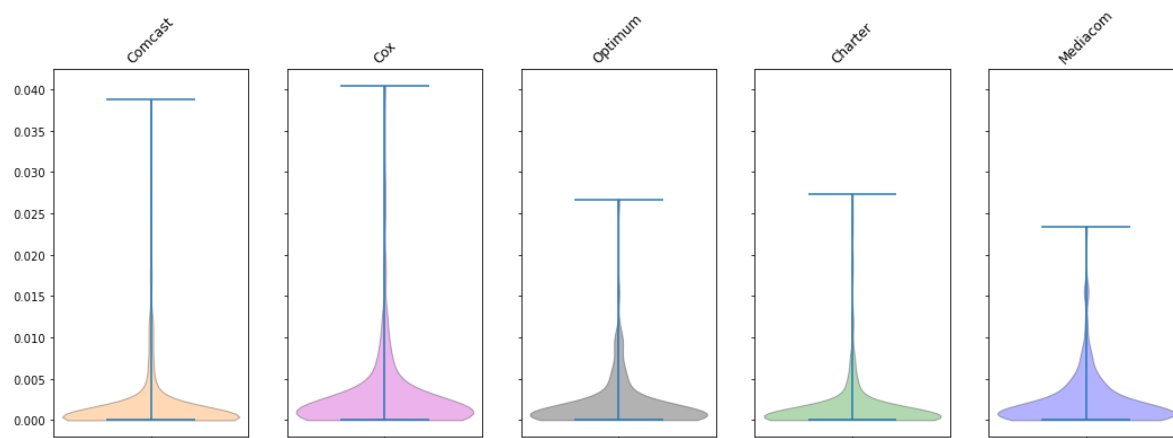


Figure A7: Distribution of losses for units attached to cable access technology, by provider.

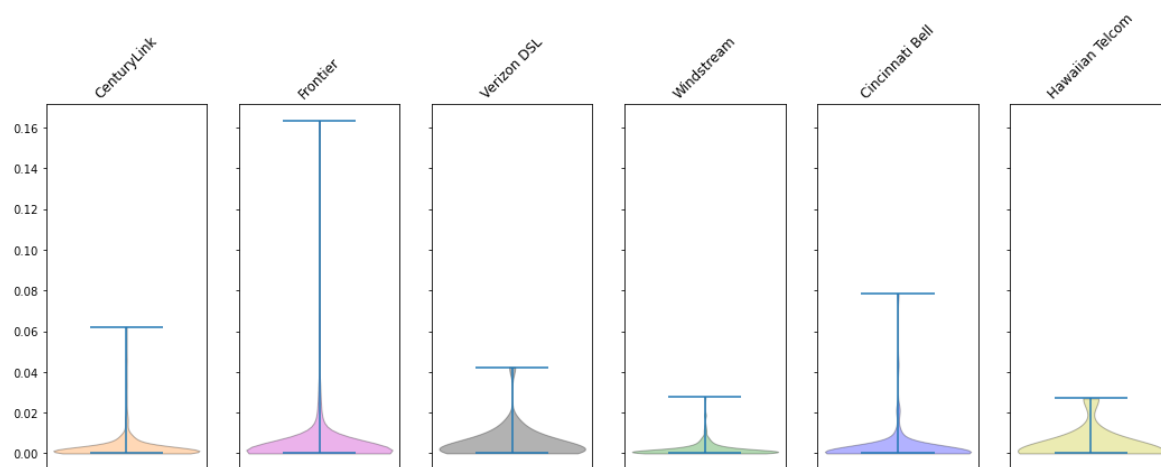


Figure A8: Distribution of losses for units attached to DSL access technology, by provider.

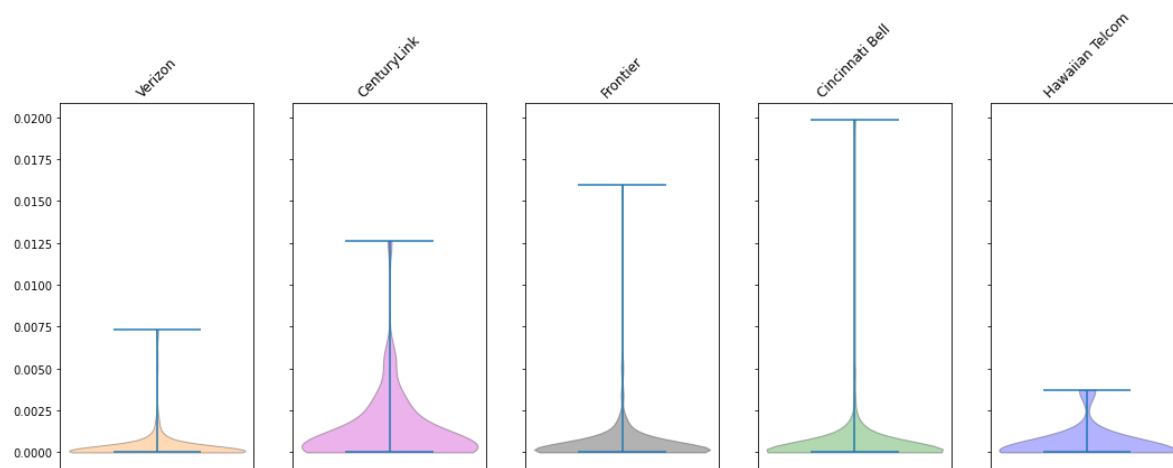


Figure A9: Distribution of losses for units attached to fiber access technology, by provider.

We were concerned with how much change there was, month to month, in the units that showed a high loss level. Were there units that had a persistent high level of loss, or did the units with high loss change from month to month? In figure A10(a) we plot the relationship between the units that were active for two consecutive months. There are a few units with a similar loss rate for the two months (those that lie along the diagonal). As well there are units with a high lost rate for one month and a much lower rate for the other. When we compare the latest data from the FCC (3/2022) with data from 8/2020 (the month we used to compare to the last validated month from the FCC), there is little visual evidence of links with similar high loss rates.

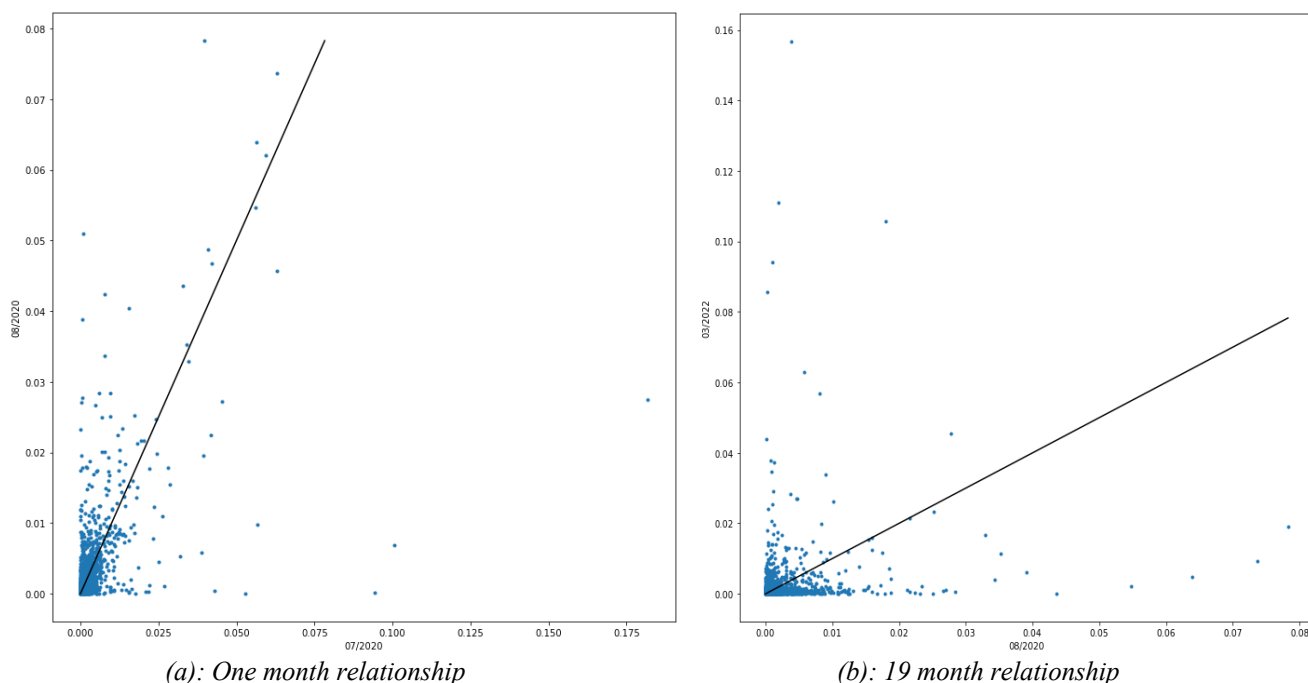
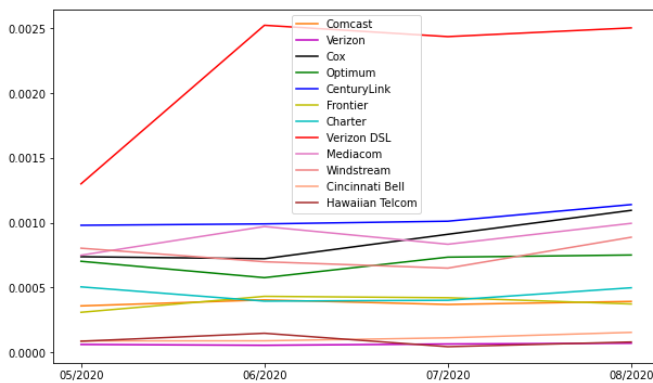
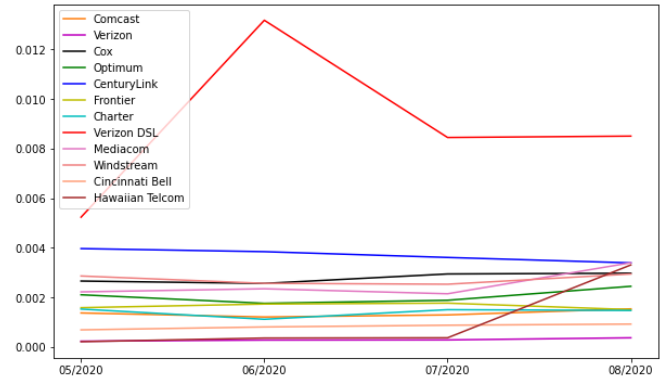


Figure A10: relationship of loss rate for different MBA units in different months. Between two adjacent months, there are units that show somewhat similar loss rates. Between 8/20 (the month we cleaned to compare to the latest validated month) and the latest month posted (3/22), there is little visible similarity in units with high loss rates.

We were interested in how measured loss rates change from month to month. We picked two definitions of the loss metric, the median loss rate across the units (50% of the units had a lower loss rate) and the maximum loss rate for 80% of the units. A definition based on an 80% cutoff, or a median value (50% cutoff) would be less sensitive to the number of high-loss units included in the samples, compared to the 99th percentile. In Figure A11 we plot the loss rate for the unit at the median and at the 80% decile of the units. There is variation among the ISPs, as well as variation across the months. In Figure A12 we focus on cable technology in particular. We suggest two conclusions from this data. First, the loss rates change from month to month, so a number from a single month may not be representative. Second, these loss rates are low enough (both median and 80%) that the differences among the providers is likely to have an effect on user QoE. Application-specific research might reveal cases where these differences would matter, but lacking that evidence, all of these numbers are below what is a realistic SSL. A “less is better” competition in this context would not have merit. We close with our reminder that you get what you measure for.

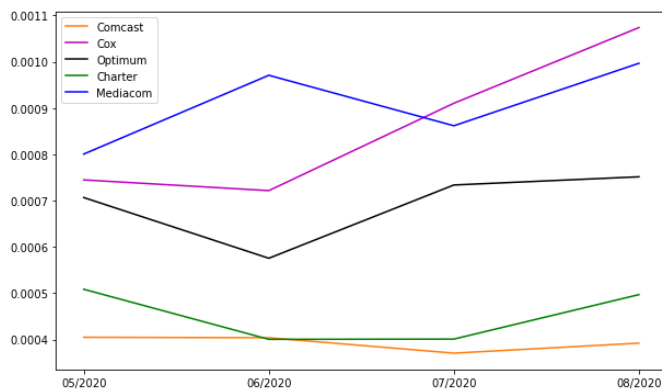


(a) :Median loss across units

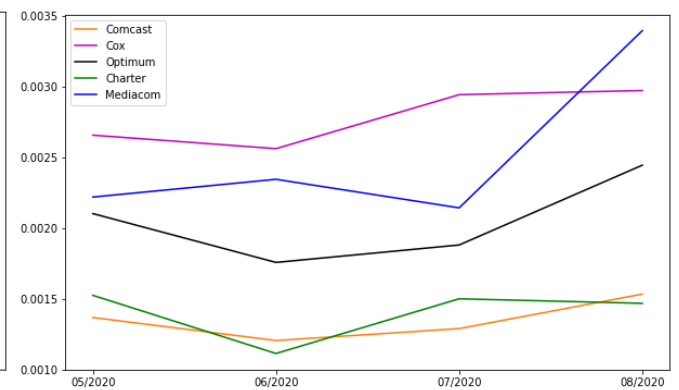


(b): Maximum loss rates for 80% of units

Figure A11: Packet loss rates for ISPs in the MBA program for four months in 2020. Verizon DSL had consistently high loss rates compared to others. There is fluctuation in the values across the months, with the ordering of some of the ISPs changing.



(a): Median loss across units



(b):Maximum loss rate for 80% of units

Figure A12: Packet loss rates for ISPs using cable technology in the MBA program.